

DAYANANDA SAGAR ACADEMY OF TECHNOLOGY AND MANAGEMENT

An Autonomous Institute under VTU, Belagavi

Accredited by NAAC with Grade A+



Engineering Exploration Course (BEEXC208)

Project Report

on

**Title: AI and Laser-Based Smart Animal Intrusion Detection and
Security System**

SDG Theme: Smart and Sustainable Agriculture

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In partial fulfillment of the requirement for the degree of

BACHELOR OF ENGINEERING

Under the Guidance of

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This is to satisfy that the **Engineering Exploration Course** project work entitled **Title: AI and Laser-Based Smart Animal Intrusion Detection and Security System** under the **SDG Theme: Smart and Sustainable Agriculture** carried out by **Lahari S (1DT25IS079), Modini Siri S (1DT25IS089) , Nithin N (1DT25IS096) and Puneet (1DT25IS110)**, a bonafide student of **DAYANANDA SAGAR ACADEMY OF TECHNOLOGY AND MANAGEMENT** in Bachelor of Engineering of Visvesvaraya Technological University, Belagavi during the academic year 2025-2026. It is certified that all the corrections/suggestions indicated for internal assessment have been incorporated in the report. The project report has been approved as it satisfies the academic requirement in respect of the Engineering Exploration Course prescribed for the degree.

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Sincerely,

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ABSTRACT

Agriculture plays a vital role in ensuring food security and supporting the economy; however, crop damage caused by animal intrusion remains a major challenge for farmers. Animals such as monkeys, wild pigs, deer, birds, and elephants frequently enter agricultural fields, resulting in significant crop losses and economic damage. Conventional methods such as manual surveillance, fencing, and traditional scarecrows are often ineffective, labor-intensive, and unable to provide continuous monitoring. Existing systems generally rely on simple motion detection or alarm mechanisms and lack the capability to intelligently identify animals and provide timely notifications. Therefore, there is a need for an automated, intelligent, and cost-effective system that can accurately detect animal intrusion, alert farmers in real time, and improve farm security while reducing human effort.

The proposed project, “AI and Laser Based Smart Animal Intrusion Detection and Security System,” addresses these limitations by integrating Artificial Intelligence, Internet of Things (IoT), and Embedded Systems technologies. The system consists of a Laser-LDR pair, a PIR motion sensor, an ESP8266 NodeMCU microcontroller, a camera, and a buzzer. The Laser-LDR pair is used to detect perimeter intrusion, while the PIR sensor identifies motion within the monitored area. Simultaneously, a camera captures visual data that is processed using the YOLOv8 deep learning model implemented in Python for real-time animal detection and classification. The system is capable of recognizing animals such as elephants, cows, birds, sheep, deer, monkeys, and pigs. Upon detection of intrusion or animal presence, alerts are generated and transmitted to the farmer through Telegram notifications. The ESP8266 communicates with the Python application through serial communication, enabling coordinated operation between the sensor-based and AI-based modules. A buzzer is activated to provide an immediate local warning whenever an intrusion or animal detection event occurs.

The developed prototype successfully demonstrated accurate animal detection, perimeter monitoring, motion sensing, real-time alert generation, and remote notification capabilities. The integration of AI-based image processing with IoT-enabled sensing significantly improves monitoring efficiency compared to conventional approaches and minimizes the need for continuous human supervision. The system provides an affordable and scalable solution for agricultural field protection and can help reduce crop losses caused by animal intrusion. Future enhancements may include the addition of automated deterrent mechanisms such as intelligent scarecrow movement, flashing lights, animal-specific sound systems, solar-powered operation, cloud-based monitoring, and mobile application integration. These improvements can further increase the effectiveness, reliability, and practicality of the proposed system for large-scale smart farming applications.

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Abbreviations

AI	Artificial Intelligence
IoT	Internet of Things
PIR	Passive Infrared Sensor
LDR	Light Dependent Resistor
CNN	Convolutional Neural Network
YOLOv8n	You Only Look Once version 8 Nano
GSM	Global System for Mobile Communications
ESP32	Espressif Systems 32-bit Microcontroller
ESP8266	Espressif Systems 8266 Wi-Fi Microcontroller
USB	Universal Serial Bus
LED	Light Emitting Diode
SG90	9g Micro Servo Motor Model
API	Application Programming Interface
GPIO	General Purpose Input/Output
IEEE	Institute of Electrical and Electronics Engineers
SVM	Support Vector Machine
VGG	Visual Geometry Group

CHAPTER 1

INTRODUCTION

CHAPTER 1

INTRODUCTION

1. INTRODUCTION

Agriculture is a fundamental sector that supports food production and contributes significantly to economic growth. One of the major challenges faced by farmers is crop damage caused by animal intrusion. Animals such as monkeys, deer, wild pigs, birds, and elephants frequently enter agricultural fields in search of food and water, resulting in considerable crop loss and financial burden. Traditional methods of protection, including manual surveillance, fencing, and conventional scarecrows, often require continuous human intervention and may not provide effective protection. With the advancement of Artificial Intelligence (AI), Internet of Things (IoT), and Embedded Systems, it is possible to develop intelligent monitoring systems capable of detecting animal intrusion and providing timely alerts to farmers.

The proposed project, “AI and Laser Based Smart Animal Intrusion Detection and Security System,” aims to provide a smart and automated solution for protecting agricultural fields. The system combines sensor-based monitoring using a Laser-LDR pair and PIR sensor with AI-based animal detection using the YOLOv8 model. The Laser-LDR pair detects perimeter intrusion, while the PIR sensor identifies motion within the monitored area. Simultaneously, a camera captures visual data, which is analyzed by the YOLOv8 deep learning model to identify animals such as elephants, cows, birds, sheep, deer, monkeys, and pigs. Upon detection, the system generates real-time Telegram notifications and activates a buzzer to alert farmers immediately.

The integration of AI and IoT technologies enhances the reliability and effectiveness of field monitoring by enabling continuous surveillance and rapid response. The proposed system reduces dependency on manual monitoring, improves crop protection, and provides a cost-effective solution for modern agriculture. Furthermore, the system can be extended in the future with automated deterrent mechanisms, cloud-based monitoring, and renewable energy integration, making it suitable for large-scale smart farming applications.

1.1 ORGANIZATION OF THE REPORT

Organisation of the report is explained as follows:

Chapter 1 presents the introduction to the project and provides an overview of the proposed system. Chapter 2 describes the Design Thinking Approach, including customer research, empathy mapping, customer journey mapping, and problem identification. Chapter 3 presents the literature survey, prior art search, product survey, research gap, objectives, and project planning activities. Chapter 4 discusses the feasibility analysis of the proposed system from technical, financial, market, and competitor perspectives. Chapter 5 explains the methodology and implementation, including the system architecture, flowchart, performance parameters, hardware requirements, software requirements and circuit design. Chapter 6 presents the experimental results, validation, performance analysis, graphs, and prototype demonstration. Chapter 7 provides the conclusion of the project and discusses possible future enhancements. Chapter 8 Summarizes the learning outcomes, project experience, and key technical and professional skills gained during the development of the proposed system.

CHAPTER 2

DESIGN THINKING APPROACH

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DESIGN THINKING APPROACH

2.1 INTRODUCTION

Design Thinking is a human-centered, creative, and iterative problem-solving approach used to identify real-world problems and develop innovative solutions. In Innovation and Design Thinking (IDT), this approach helps students understand user needs, generate ideas, create prototypes, and test solutions effectively. Unlike traditional methods that focus only on technical aspects, Design Thinking emphasizes empathy, experimentation, collaboration, and continuous improvement.

The Design Thinking process encourages individuals to view problems from the user's perspective and develop solutions that are practical, feasible, and desirable. It combines analytical thinking with creativity, enabling innovators to address complex challenges in fields such as healthcare, agriculture, education, technology, and environmental sustainability.

In today's rapidly changing world, Design Thinking plays a vital role in fostering innovation. It helps organizations and individuals create products, services, and systems that meet user expectations while addressing social and economic needs. By promoting teamwork, critical thinking, and problem-solving skills, Design Thinking has become an essential framework for innovation and entrepreneurship.

Stages of Design Thinking

2.1.1 Empathize

- Understand the needs, experiences, and challenges of users.
- Gather information through observation, interviews, and surveys.

2.1.2 Define

- Clearly identify and define the problem based on user insights.
- Frame the problem statement accurately.

2.1.3 Ideate

- Generate multiple creative ideas and possible solutions.
- Encourage brainstorming without limitations.

2.1.4 Prototype

- Develop a basic model or representation of the selected solution.
- Create low-cost and simple prototypes for testing.

2.1.5 Test

- Evaluate the prototype with users.
- Collect feedback and refine the solution for better performance.

Design Thinking is a powerful methodology that bridges the gap between identifying a problem and creating an effective solution. In IDT, it serves as a structured framework that enables innovators to develop user-friendly, sustainable, and practical solutions. By following the stages of Empathize, Define, Ideate, Prototype, and Test, individuals can transform ideas into successful innovations that create meaningful value for society.

2.2 SUSTAINABLE DEVELOPMENT GOALS

Zero Hunger : Goal 2 aims to end hunger, achieve food security, improve nutrition, and promote sustainable agriculture. Hunger and malnutrition remain major challenges in many parts of the world, affecting the health and development of millions of people. This goal focuses on ensuring that everyone has access to sufficient, safe, and nutritious food throughout the year. It also emphasizes increasing agricultural productivity, supporting small-scale farmers, and adopting sustainable farming practices that protect the environment and natural resources for future generations.

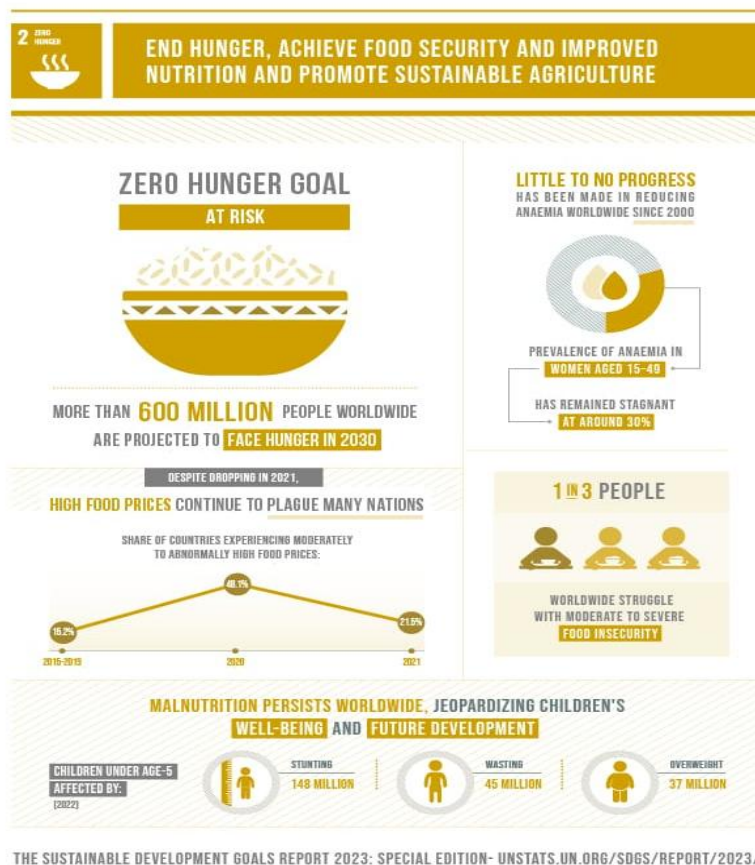


Fig 2.1 – SUSTAINABLE DEVELOPMENT GOAL 2: ZERO HUNGER

SDG 2 seeks to eliminate hunger and all forms of malnutrition while ensuring sustainable food production systems. It encourages the use of modern technologies and innovative farming methods to increase crop yield and reduce food wastage. Regions facing food shortages, climate challenges, and poor agricultural infrastructure are given special priority.

The theme of Smart and Sustainable Agriculture aligns closely with SDG 2 by promoting the use of advanced technologies such as sensors, IoT devices, artificial intelligence, and automated monitoring systems in farming. These technologies help farmers make informed decisions, optimize resource usage, improve crop productivity, and minimize environmental impact. Smart agriculture contributes to food security while ensuring the long-term sustainability of agricultural practices, supporting the vision of a hunger-free world.

2.3 USER RESEARCH



Fig 2.2 – PERSONA 1: SENIOR VETERINARY OFFICER

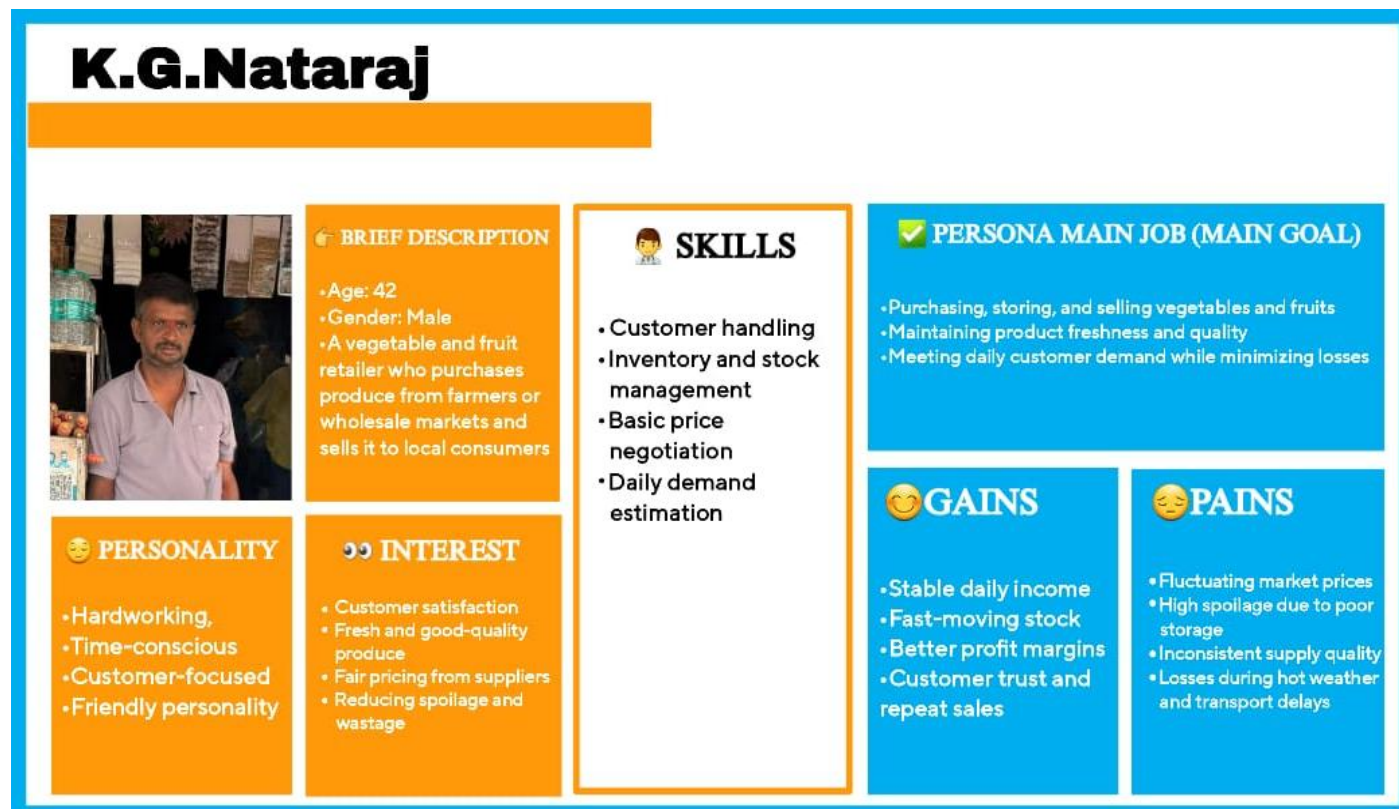


Fig 2.3 – PERSONA 2: VEGETABLE RETAILER

Shrinivas



BRIEF DESCRIPTION

- Gender: Male
- Age: 55
- A farmer cultivates crops and/or raises livestock to produce food and raw materials for society.



SKILLS

- Machinery handling
- Problem-solving
- Basic finance
- Risk management
- Crop planning
- Irrigation

PERSONA MAIN JOB (MAIN GOAL)

- Growing crops and managing livestock from sowing to harvesting and selling produce



PERSONALITY

- Hardworking
- patient
- Resilient Practical
- Close to nature

INTEREST

- Farming techniques
- New seeds/technology
- Market prices
- Weather patterns
- Soil health



GAINS

- Food production
- Income from crops
- Self-reliance
- Satisfaction of feeding society



PAINS

- High input costs
- Water scarcity
- Unpredictable weather
- Low market prices, debt

Fig 2.4 – PERSONA 3: SMALL-SCALE FARMER

Manjunath



BRIEF DESCRIPTION

- Gender: Male
- Age: 37
- A consumer intakes the fresh or less chemical free foods like vegetables and fruits.



SKILLS

- Time management
- Problem solving

PERSONA MAIN JOB (MAIN GOAL)

- Currently working in IT field with 6-8 years experience



PERSONALITY

- patient
- Time conscious
- Detail oriented

INTEREST

- Sustainable farming practices
- Traceability (knowing where food comes from)
- Organic and chemical-free food



GAINS

- work life balance
- stable Income
- Reliable
- Opportunity



PAINS

- High cost and value doubts
- Lack of trust and authenticity
- Rapidly changing technologies
- Poor and unclear product information

Fig 2.5 – PERSONA 4: COMMERCIAL FARMER

2.4 EMPATHY MAP

Empathy Mapping is a user-centered design thinking tool used to gain a deeper understanding of the needs, experiences, emotions, and challenges of the end user. It helps designers analyze what a user says, thinks, does, and feels while interacting with a product or facing a particular problem. By capturing these perspectives, empathy mapping enables the development of solutions that are more relevant, effective, and user-friendly. In the context of our Smart Animal Intrusion Detection System, empathy mapping helps us understand the concerns of farmers, such as crop losses due to animal intrusions, the difficulty of monitoring fields continuously, and the need for timely alerts. By identifying their frustrations, expectations, and daily experiences, we were able to design a smart, reliable, and affordable system that addresses their real-world agricultural challenges and improves field security.

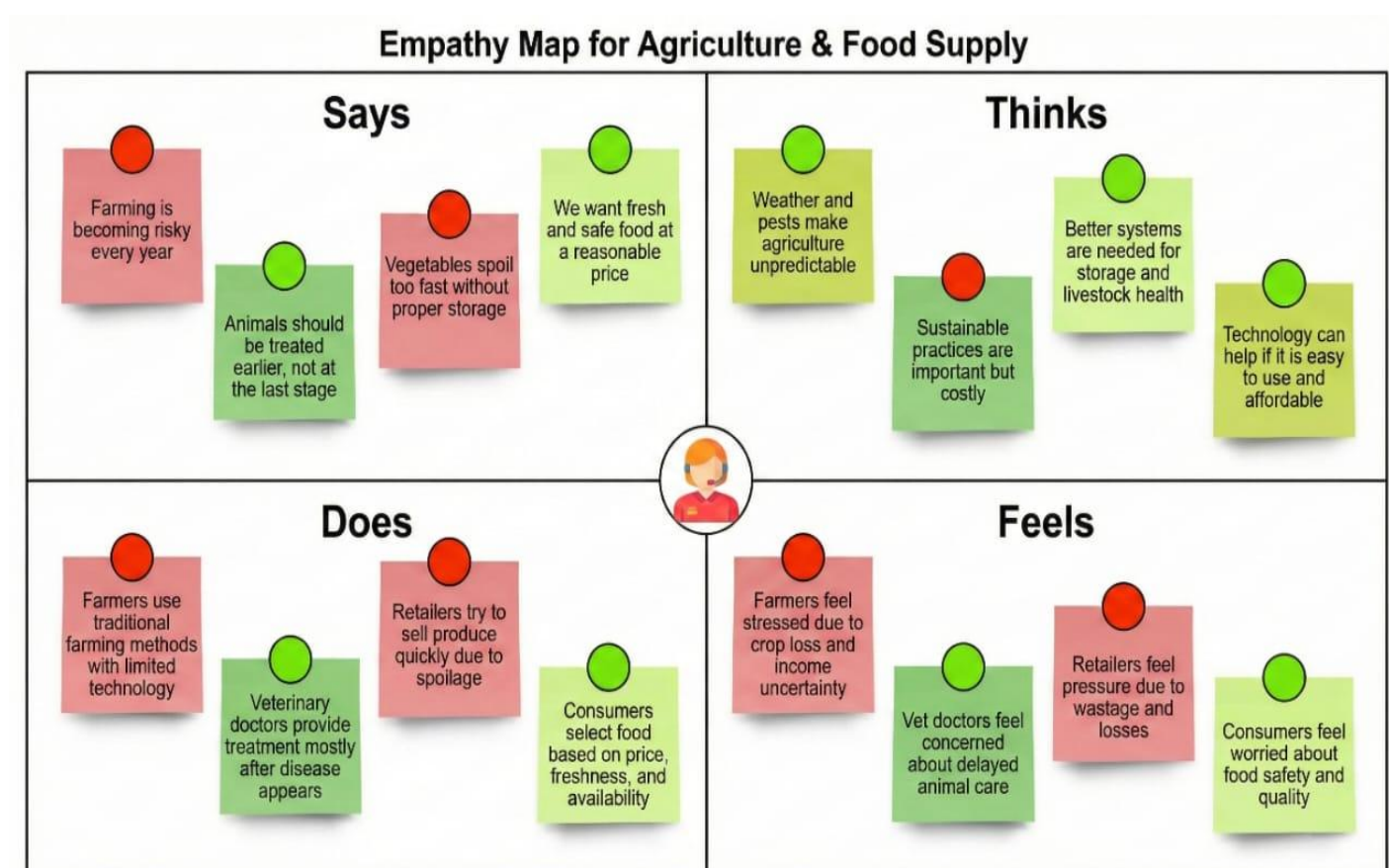


Fig 2.6 – Empathy Map for Agricultural Field Users

2.5 CUSTOMER JOURNEY MAP

Customer Journey Mapping is a design thinking tool used to visually represent the complete experience of a user while interacting with a product, service, or system. It helps identify the various stages a customer goes through, from initial awareness to final usage and feedback, while highlighting their actions, needs, emotions, challenges, and expectations at each stage. By analyzing these touchpoints, designers and developers can better understand user behavior, identify pain points, and discover opportunities for improvement. In the context of our Smart Animal Intrusion Detection System, customer journey mapping helps us understand the farmer's experience—from recognizing the problem of crop damage, seeking an effective solution, receiving real-time intrusion alerts, responding to warnings, and ultimately benefiting from improved crop protection. This approach ensures that the developed solution is user-centered, practical, and aligned with the actual needs of farmers.

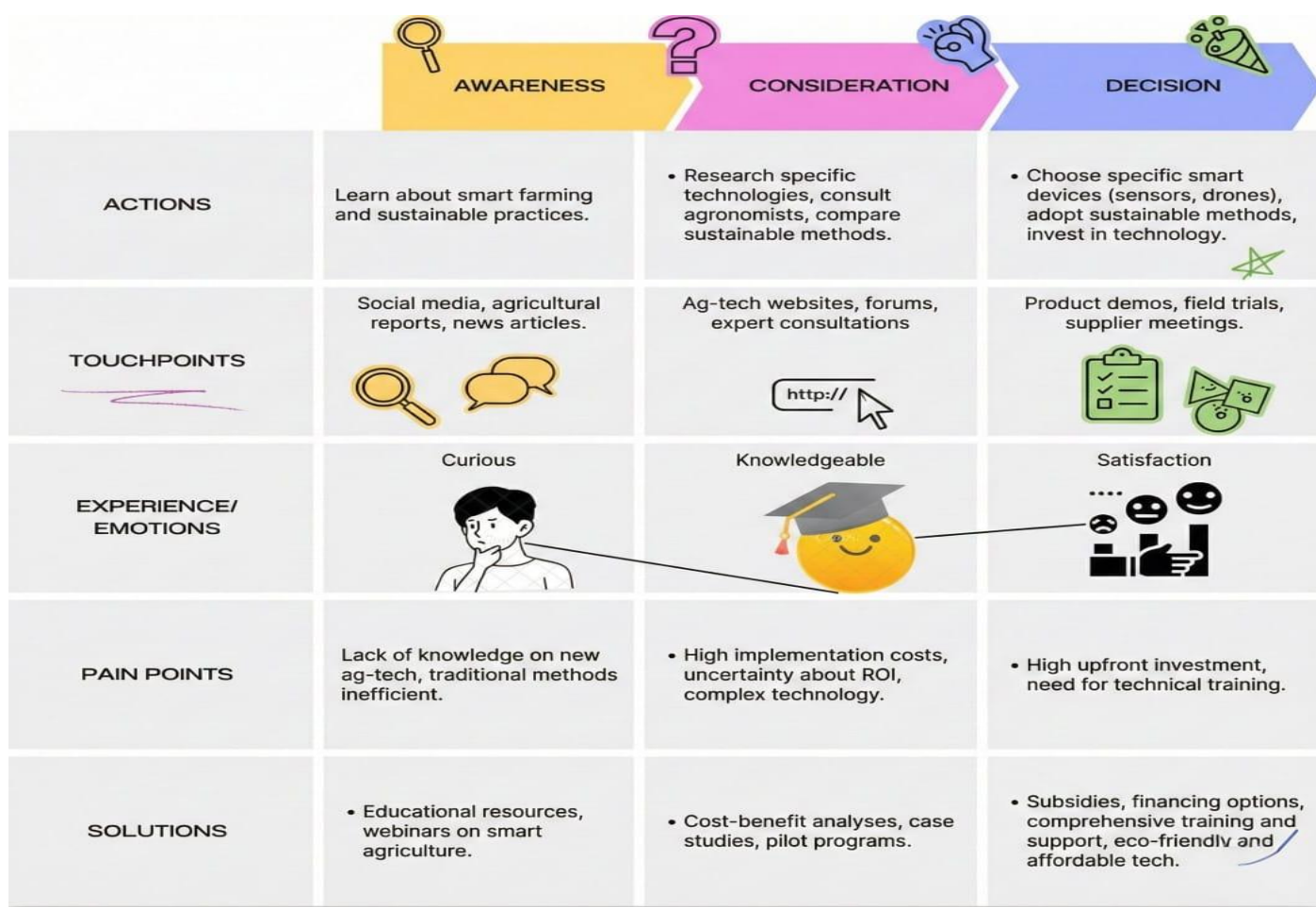


Fig 2.7 – Customer Journey Map for Smart Animal Intrusion Detection System

2.6 NEED STATEMENT TO PROBLEM DEFINITION

Agriculture plays a vital role in ensuring food security and supporting the livelihoods of millions of people. However, farmers face numerous challenges such as crop damage caused by animals, pests, unpredictable weather conditions, and inefficient monitoring of agricultural fields. These issues often lead to reduced crop yield, economic losses, and increased pressure on food production systems.

Secondly, traditional farming methods rely heavily on manual supervision, which can be time-consuming and ineffective, especially for large agricultural lands. Farmers may not always be present to monitor their fields, making crops vulnerable to animal intrusion and other threats. The lack of timely detection and response mechanisms further increases the risk of crop damage and resource wastage.

Lastly, the growing demand for food, combined with limited agricultural resources, highlights the need for smart and sustainable farming solutions. Advanced technologies such as sensors, artificial intelligence, image processing, and automated alert systems can help farmers monitor their fields in real time and take immediate action when threats are detected. These factors collectively emphasize the need for an intelligent crop protection system that enhances agricultural productivity, reduces losses, and supports sustainable farming practices while contributing to food security and the achievement of SDG 2: Zero Hunger.

- Financial losses faced by farmers because of crop destruction.
- Need for early detection and timely alerts to prevent damage.
- Growing demand for higher agricultural productivity and food security.
- Requirement for efficient utilization of water, fertilizers, and other resources.
- Need to reduce human effort through automation and smart technologies.
- Importance of adopting sustainable farming practices to protect the environment.
- Necessity of integrating modern technologies such as IoT, sensors, AI, and image processing in agriculture.
- Need to improve crop yield and overall farm management.

CHAPTER 3

LITERATURE SURVEY

CHAPTER 3

LITERATURE SURVEY

3.1 INTRODUCTION

Human–wildlife conflict has become a major challenge in agricultural regions, causing significant crop damage, economic losses, and safety concerns for farmers. Wild animals such as elephants, monkeys, wild pigs, deer, cattle, and birds frequently enter farmlands in search of food and water, resulting in destruction of crops and property. Traditional protection methods such as fences, scarecrows, and manual monitoring are often ineffective, labor-intensive, and unable to provide continuous surveillance.

Recent advancements in the Internet of Things (IoT), Wireless Sensor Networks (WSN), Computer Vision, and Artificial Intelligence (AI) have enabled the development of automated animal intrusion detection systems. Several researchers have proposed solutions using PIR sensors, ultrasonic sensors, cameras, GSM modules, and IoT platforms to detect animal movement and send alerts to farmers. Sensor-based systems provide low-cost intrusion detection but often fail to accurately identify the type of animal. Image processing and machine learning techniques have improved detection accuracy by classifying animals from captured images and videos. More recently, deep learning models such as CNN, YOLO, VGG, and Bi-LSTM have demonstrated high accuracy in real-time animal detection and monitoring applications.

The work by Prajna P. et al. proposed an IoT-based wild animal intrusion detection system that combines PIR sensors, cameras, image processing, GSM communication, and repellent mechanisms such as bright lights and sound generators to protect agricultural fields. Other studies have utilized IoT-enabled sensor networks and machine learning techniques to detect intrusions and provide notifications to farmers through mobile applications and cloud platforms. Advanced deep learning approaches have further improved detection performance by employing CNN, YOLO, VGG-19, and Bi-LSTM architectures for accurate animal classification and alert generation.

Although existing systems successfully detect animal intrusion and generate alerts, many of them primarily focus on monitoring and notification rather than actively preventing crop damage. Most systems lack intelligent deterrent mechanisms capable of responding differently to various animal species. Therefore, there is a need for an integrated solution that combines IoT-based sensing, AI-powered animal recognition, real-time alert generation, and automated deterrent actions.

Motivated by these limitations, the proposed project, “AI and Laser Based Smart Animal Intrusion Detection and Security System,” integrates Laser-LDR intrusion sensing, PIR-based motion detection, YOLOv8-based animal classification, Telegram notifications, and automated deterrent mechanisms such as buzzers, flashing lights, and a servo-based moving scarecrow. This approach not only detects and identifies intruding animals but also initiates appropriate preventive actions to reduce crop damage and enhance agricultural field protection.

3.2 SURVEY OF PAPER

Summary of existing works is given in the table number 3.1 as follows

Table 3.1 Survey of Existing Research Papers on Animal Intrusion Detection

SL NO	REFERENCE	PROBLEM ADDRESSED	COMPONENTS USED	TECHNIQUE USED	PARAMETERS USED	RESULT	ADVANTAGE	DIS-ADVANTAGE
1	IoT-based Wild Animal Intrusion Detection System” by Prajna P., Soujanya B.S, Mrs. Divya	Preventing wild animal intrusion in agricultural fields and reducing crop damage and human-animal conflict	PIR Sensors, Camera, Microcontroller, GSM Module, Light Emitter, Sound Generator, PC/Image Processing Unit, Solar Panel, Battery	IoT-based monitoring with image processing and animal classification	PIR sensor range: 30 m, Camera range: 50 m, Image matching score (0–1), GSM alert system	Detects animal movement, captures image, classifies animal, activates repellent system, and sends SMS alerts to farmers and forest officials	Reduces crop loss, provides automatic monitoring, minimizes false alarms, eco-friendly alternative to electric fencing	Requires continuous power supply, image processing complexity, limited accuracy in poor weather/light conditions
2	“Wild Animal Activity Detection Using Hybrid Deep Neural Networks” by Nomula Sowmya, Dr. B. Srinivas Rao	Detecting wild animals, monitoring their activity, and sending alert messages to prevent human-animal conflicts and hunting	Surveillance Cameras, Drones, Hybrid VGG-19 + Bi-LSTM Network, SMS Alert System, Benchmark Datasets	Deep Learning, CNN, Hybrid VGG-19 + Bi-LSTM, Image Classification and Activity Detection	40,000 images, 25 classes, 98% accuracy, 77.2% mAP, 170 FPS	Successfully detects and classifies wild animals, monitors activities, and sends SMS alerts to forest officers	High accuracy, fast processing, automatic alert generation, improves human and animal safety	Requires high GPU resources, complex implementation, large dataset requirement
3	“A Deep Learning Approach to Detecting and Deterring Birds Integrated with a Laser System” by Muhammad Sajjad et al.	Preventing crop damage caused by birds and reducing financial loss to farmers	Camera, YOLOv11 Deep Learning Model, Jetson TX2 Embedded System, Laser Control Unit, 400-mW, Relay, Motor Control Chip, Ubuntu 18.04, Python	Deep Learning-based Bird Detection and Laser Deterrence using YOLOv11	Accuracy: 96.02%, Macro Precision: 95.87%, Macro Recall: 95.46%, Image Resolution: 1920×1080, IoU Threshold: 0.1–0.5, Confidence Level: 0.90–0.95	Successfully detects and deters birds automatically using rotating laser mechanism with high accuracy	High detection accuracy, automatic operation, eco-friendly, detects even small birds, reduces crop loss without harming birds	Requires high computational resources, depends on camera quality and lighting conditions, complex system setup

4	“Protection of Crop Fields from Grazing Animals by Using Laser Beam Technology” by Sravanthi Pammi S.S., B. Raja Rajeswari, G. Madhu Babu, S.	Protecting crop fields from grazing animals and reducing crop damage	Laser Light, LDR Sensor, BC547 Transistor, Buzzer, 2.2 K Ω Resistor, 9V Battery, Connecting Wires	Laser Beam Interruption Detection System	Laser wavelength: 630–670 nm, 9V DC supply, LDR light sensitivity	Detects interruption caused by animals and activates alarm sound to scare them away	Low cost, easy to implement, low power consumption, reduces manpower requirement	Limited coverage area, false alarms possible, depends on proper laser alignment
5	Authors et al., 2022 (Yolov5)	Detecting animal intrusion in farms to prevent crop damage and safety risks	Camera / Image input system Computer (GPU/CPU for processing) YOLOv5 model (software) Dataset of animal images Deep learning framework (like PyTorch)	Deep Learning (Computer Vision) and YOLOv5 and Backbone: CSP (Cross Stage Partial) Network	mAP (mean Average Precision) Accuracy and Detection performance (speed + reliability)	Achieved ~94% mAP (high accuracy) Effective real-time animal detection Improved detection performance significantly	High accuracy detection Real-time monitoring Reduces crop damage Helps prevent human-animal conflict Automated system (less manual effort)	Requires large dataset- training Performance may drop in: Low light / night conditions, Bad weather, Needs good hardware (GPU), Detects limited animal categories (4 types), No integrated alert system
6	Amol More (2025) • Journal of Advancement in Machines	• Crop damage due to wild animals. • Need for low-cost farm security	• Laser diode and LDR sensor. • Arduino UNO and buzzer	• Laser Beam Security System (LBSS) • LDR sensor with IoT and buzzer alert	• Detection accuracy. • False alarm rate	• Worked with buzzer frequency of 12–20 kHz. • Reduced false alarms and protected up to 100% crop-loss risk areas	• Low power consumption. • Easy installation and maintenance	• Laser alignment can be disturbed • Limited coverage for very large farms

7	• Prashanth C. Ravoor, T.S.B. Sudarshan, Krishnan Rangarajan (2021)	• Human-animal conflict near forest areas • Crop damage and danger to humans due to animal intrusion	• Raspberry Pi with camera module • PIR sensors, edge server and MQTT communication	• Deep Learning and Computer Vision. • MobileNetv2-SSD, Res-Net50 and Raspberry Pi	• Detection accuracy • Re-identification accuracy and false positive rate	• Detection accuracy: 80%, 89.47%, 92.56% for tiger, jaguar and elephant • Re-identification accuracy: 99.6%, 86.2%, 61.7% with 2–3 fps processing speed	• Real-time and scalable monitoring system. • Can track animals across multiple cameras	• Lower speed on embedded devices. • Alarm systems may become ineffective as animals adapt
8	• Prabhat Kumar Panda et al. (2022)	• Crop damage caused by wild animal intrusion. • Human safety risks near forest farming areas	• Arduino UNO, ESP32-CAM and ultrasonic sensors • Servo motor, NRF24L01 module and buzzer	• IoT-based intrusion detection system • Ultrasonic sensors with ESP32-CAM and E-vehicle	• Intrusion detection performance. • Image capture and alert notification efficiency	• Ultrasonic sensor range: 2–400 cm. • Servo motor operating speed: 0.12 sec with successful real-time alerts	• Provides automatic alert notification to farmers. • Low-cost and real-time monitoring system	• Limited coverage area of sensors. • Future improvements needed using IR and wide-angle cameras
9	Authors et al., 2023 – Wild Animal Intrusion Detection in Fields Using IoT	Crop damage and farmer safety issues caused by wild animal intrusion in agricultural fields	IoT sensors, Camera module, GSM module, Microcontroller, Alarm system	IoT-based animal detection and alert system	Accuracy, Detection speed, Response time	Successfully detected animal movement and generated instant alerts to farmers	Real-time monitoring, reduces crop damage, automated alerts	Requires stable network connection and regular maintenance
10	Authors et al., 2024 Detection of Animal Intrusion in Agricultural Field Using RNN	Difficulty in detecting animals entering fields during day and night conditions	Camera sensors, Processor/GPU, Dataset, IoT modules	Recurrent Neural Network (RNN) with Deep Learning	Precision, Recall, Accuracy, Efficiency	Achieved high detection accuracy with improved classification performance	Better detection accuracy, automatic monitoring, less human effort	High computational requirement and dependency on quality dataset

11	Authors et al., 2025 – Recent Trends and Challenges in Animal Intrusion Detection	Need for efficient and smart monitoring systems to prevent wildlife intrusion into farms	Cameras, Sensors, Embedded systems, AI-based monitoring devices	CNN and Deep Learning based intrusion detection	Accuracy, F1-score, Detection efficiency	AI models improved real-time detection and reduced false alarms	Smart automation, continuous monitoring, improved security	Expensive hardware setup and reduced efficiency in poor lighting
12	Jeevitha S. et al., 2020 (IJERT)	Human-animal conflict in agriculture fields causing crop damage and danger to human life	IoT sensors, GSM module, Camera, Raspberry Pi, Arduino Uno, PIR sensor, Motion sensor	Image Processing, IoT-based Animal Intrusion Detection, CNN, SVM, Wireless Sensor Network	Accuracy, Detection efficiency, Energy consumption, Classification performance	Achieved effective real-time animal detection and alert notification using sensors and image classification	Real-time monitoring, automatic SMS alerts, low energy consumption, reduces crop damage and human effort	Depends on image quality and sensors, requires stable network connectivity, lower accuracy in complex environments
13	Vijayshri Vaidya et al., 2025	The need for a rapid, real-time automated system to detect plant diseases in the field to prevent significant economic losses and reduce the heavy reliance on human experts.	OpenCV, MobileNet model, IoT devices, Camera sensors, WhatsApp alerts, Loudspeaker system	Deep Learning, Computer Vision, MobileNet, IoT-based monitoring	Accuracy, Precision, Recall, Detection efficiency	Real-time wild animal detection and automatic alert generation with improved monitoring	Automated detection, real-time alerts, reduces crop damage, protects both humans and animals	Requires proper dataset training, internet connectivity, and good camera visibility
14	D. Archana et al., 2023	Wild animal intrusion into agricultural fields causing crop damage and safety issues.	ESP32 Camera, IoT modules, Sensors, GSM/Wi-Fi modules, OpenCV, CNN models	Machine Learning, CNN, IoT-based intrusion detection system	Accuracy, Detection speed, Classification performance	High accuracy animal detection with automatic alert notifications to farmers	Real-time monitoring, automated alerts, improves farm security, reduces manual effort	Needs large training dataset, performance affected in low light and bad weather conditions

15	Garg, R., et al., "Real-Time Plant Disease Detection Using Edge AI on Plantify's IoT Network,"	The challenge of early plant disease diagnosis to minimize crop loss and ensure food security, specifically moving away from manual inspection to automated real-time monitoring.	ESP32 microcontroller, DHT22 (temperature and humidity) sensor, LDR (light intensity) sensor, soil moisture sensors, image capturing cameras, and a cloud-based software platform.	An IoT-based monitoring system using Edge AI and Machine Learning (specifically Convolutional Neural Networks - CNN) for both sensor data analysis and image processing.	Temperature, humidity, light intensity, soil moisture levels, and plant image features for classification.	The system successfully demonstrated real-time monitoring of health parameters and accurate detection of diseases with automated alerts and treatment recommendations.	Enables immediate intervention through real-time data analysis; low-cost, user-friendly, and reduces the need for manual field presence.	Currently requires high-quality image datasets for training and its applicability is initially limited to specific plant species before further scaling.
16	M. Khalid et al., "Palm Leaf Health Management: A Hybrid Approach for Automated Disease Detection and Therapy Enhancement,"	The significant impact of hazardous plant leaf diseases on crop quality and productivity; specifically, the difficulty in accurately and quickly distinguishing between healthy and damaged palm leaves in agricultural settings.	Digital camera/imaging sensors for dataset acquisition, Efficient Channel Attention Network (ECA-Net), ResNet50, and DenseNet201 architectures.	A novel hybrid deep learning model combining ECA-Net with transfer learning (fusion of ResNet50 and DenseNet201) and computer vision.	Training accuracy, validation accuracy, F1-score, and precision metrics.	The hybrid model achieved an outstanding 99.54% training accuracy and maintained a 98.67% validation accuracy.	High precision in disease diagnosis and superior performance compared to earlier models; it sets a new benchmark for F1-score values in agricultural technology.	As a complex hybrid deep learning model, it may require significant computational power for real-time deployment compared to simpler architectures.

3.3 PRIOR ART SEARCH

The prior art review indicates that most existing animal intrusion detection systems can be classified into three categories. Sensor-based systems use PIR, ultrasonic, laser, and vibration sensors to detect movement but cannot identify the exact animal species. IoT-based alert systems employ GSM, Wi-Fi, or cloud platforms to provide real-time notifications; however, they generally do not prevent intrusion after detection. AI and Deep Learning-based systems utilize CNN, YOLO, VGG, Bi-LSTM, and related models for animal classification, offering higher accuracy but often lacking integrated deterrent mechanisms and requiring higher computational resources.

The analysis reveals that most existing solutions focus mainly on detection and notification, provide limited multi-sensor integration, rarely implement animal-specific deterrent actions, and often lack real-time communication between AI software and embedded hardware.

To address these limitations, the proposed AI and Laser Based Smart Animal Intrusion Detection and Security System integrates Laser-LDR intrusion sensing, PIR motion detection, YOLOv8-based animal classification, Telegram notifications, ESP8266-Python communication, and automated deterrent mechanisms such as buzzers, flashing LEDs, and a servo-motor-driven moving scarecrow. This approach combines detection, identification, alerting, and prevention within a single intelligent agricultural protection system.

3.4 SURVEY OF PRODUCTS

Existing Commercial and Research Products

Various products and systems have been developed to reduce crop damage caused by animal intrusion. These solutions generally focus on detection, monitoring, alert generation, or deterrence. Based on the surveyed literature and existing market solutions, the following categories of products are commonly used.

3.4.1 Electric Fencing Systems

Electric fencing is commonly used to prevent animals such as elephants, wild boars, and cattle from entering agricultural fields. Although effective, installation and maintenance costs are high, and safety concerns may arise for both animals and humans.



Fig 3.1 Existing Electric Fencing System for Crop Protection

3.4.2 Conventional Scarecrow Systems

Traditional scarecrows are widely used in agricultural fields to deter birds and small animals. These systems are inexpensive and easy to deploy but remain stationary and often become ineffective as animals gradually adapt to their presence.



Fig 3.2 Traditional Scarecrow System for Animal Deterrence

3.4.3 Camera-Based Surveillance Systems

Camera surveillance systems monitor agricultural fields and record intrusion events. Farmers are often required to continuously monitor video feeds or review recordings, making the process time-consuming and less practical for large agricultural areas.



Fig 3.3 Camera-Based Surveillance System for Agricultural Fields

3.4.4 Ultrasonic Animal Repellent Systems

Ultrasonic animal repellent systems use high-frequency sound waves to deter animals from entering protected areas. These systems operate automatically and are easy to install, but their effectiveness varies depending on the animal species and environmental conditions.



Fig 3.4 Ultrasonic Animal Repellent System for Crop Protection

3.5 GAP IDENTIFIED

The literature survey and product survey reveal several limitations in existing animal intrusion detection and agricultural protection systems.

Gap 1: Focus on Detection Rather Than Prevention

Most existing systems successfully detect animal intrusion and generate alerts but do not actively prevent animals from entering or remaining in the field. Farmers still need to manually respond after receiving notifications.

Gap 2: Lack of Animal-Specific Deterrent Mechanisms

Many solutions use the same deterrent method for all animals. Different animals react differently to lights, sounds, and movements. Existing systems rarely implement species-specific deterrent actions.

Gap 3: Limited Multi-Sensor Integration

Several systems rely on only one sensing technology such as PIR sensors, ultrasonic sensors, or cameras. Single-sensor approaches can result in false detections and reduced reliability.

Gap 4: Absence of Real-Time AI and Embedded System Integration

Many AI-based solutions perform animal classification separately from embedded hardware. Real-time communication between AI software and field devices is often not implemented.

Gap 5: High Deployment Cost

Advanced surveillance systems require expensive cameras, processing units, and infrastructure, making them less affordable for small and medium-scale farmers.

Gap 6: Insufficient Automation

Most systems stop at alert generation. Automatic activation of deterrent devices such as scarecrows, warning lights, and alarms is not commonly integrated.

Gap 7: Limited Scalability for Agricultural Fields

Several research prototypes are designed for laboratory environments and are not easily scalable for large agricultural lands requiring multiple monitoring nodes.

3.6 PROBLEM STATEMENT

Agricultural fields frequently suffer from crop damage caused by wild and domestic animals such as elephants, monkeys, wild pigs, deer, birds, cattle, and sheep. Traditional protection methods including manual monitoring, fencing, and conventional scarecrows are often ineffective, labor-intensive, and unable to provide continuous surveillance. Existing animal intrusion detection systems primarily focus on sensing animal movement and generating notifications to farmers, but they provide limited support for actively preventing crop damage.

Furthermore, most available solutions lack accurate animal identification, real-time communication between Artificial Intelligence and embedded hardware, and automated species-specific deterrent mechanisms. As a

result, farmers continue to experience economic losses due to delayed response and ineffective intrusion prevention.

Therefore, there is a need for an intelligent, affordable, and scalable system that can accurately detect animal intrusion, identify the specific animal species using Artificial Intelligence, notify farmers in real time, and automatically activate appropriate deterrent mechanisms to drive animals away before significant crop damage occurs.

The proposed "**AI and Laser Based Smart Animal Intrusion Detection and Security System**" addresses this problem by integrating Laser-LDR intrusion sensing, PIR motion detection, YOLOv8-based animal recognition, Telegram-based alert notifications, ESP8266 communication, and automated deterrent mechanisms such as buzzers, flashing LEDs, and a servo-motor-driven moving scarecrow. This integrated approach enhances field protection by combining detection, identification, alerting, and prevention within a single smart agricultural security system.

3.7 DESIGN SPECIFICATIONS

Hardware Specifications

ESP8266 NodeMCU

The ESP8266 NodeMCU acts as the main controller of the system. It continuously monitors the Laser-LDR pair and PIR motion sensor, processes intrusion events, activates deterrent devices, and communicates with Telegram and the Python-based AI system.

Laser-LDR Pair

A Laser-LDR arrangement is used to create a virtual security boundary around the protected area. Any interruption of the laser beam indicates a possible intrusion and triggers the alert mechanism.

PIR Motion Sensor

The Passive Infrared (PIR) sensor detects movement of animals or humans by sensing changes in infrared radiation. It serves as a secondary detection mechanism to improve reliability and reduce false detections.

Camera Module/Web Camera

A camera is used to capture live video or images of the intruding object. The captured frames are processed by the AI model for animal identification and classification.

Buzzer

The buzzer provides an audible alert when an intrusion is detected. It acts as both a warning system and a basic animal deterrent.

LED Indicators

LEDs are used to represent flashing light deterrents. Different LED patterns can be activated depending on the type of animal detected.

Servo Motor (SG90)

The SG90 servo motor is used to implement a prototype moving scarecrow mechanism. It performs left-right motion to simulate animal-repelling movements when birds or monkeys are detected.

Personal Computer/Laptop

The laptop runs the Python application, YOLOv8 animal detection model, Telegram notification service, and serial communication interface.

Software Specifications

The system operates on Windows 10/11 and is developed using Python 3.x and Embedded C/C++ through Arduino IDE. Development tools include Arduino IDE, Visual Studio Code, and PyCharm. The AI framework used is YOLOv8 Nano (YOLOv8n). Communication is achieved through Serial Communication (USB), Wi-Fi Communication, and Telegram Bot API. The software utilizes OpenCV, Ultralytics YOLO, PySerial, Requests, and ESP8266 Wi-Fi Libraries.

Functional Specifications

Intrusion Detection

The system shall detect intrusion using a Laser-LDR pair and PIR motion sensor installed along the field boundary.

Animal Identification

The system shall identify animals such as elephants, cows, birds, sheep, deer, monkeys, and pigs using the YOLOv8 deep learning model.

Alert Generation

The system shall send instant Telegram notifications containing intrusion details and detected animal names to the farmer.

Deterrent Activation

The system shall activate appropriate deterrent devices such as buzzers, flashing LEDs, and a servo-based moving scarecrow depending on the detected animal category.

Real-Time Communication

The Python application shall communicate with the ESP8266 through serial communication to activate field-level deterrent mechanisms immediately after AI detection.

Continuous Monitoring

The system shall provide continuous monitoring of agricultural fields and operate in real time.

3.8 OBJECTIVES

Primary Objective

To develop an intelligent AI and Laser Based Smart Animal Intrusion Detection and Security System capable of detecting animal intrusion, identifying the animal species, notifying farmers in real time, and automatically activating deterrent mechanisms to prevent crop damage. The system aims to provide continuous 24×7 field surveillance by combining AI-based image recognition with laser-assisted intrusion detection for enhanced accuracy in both daytime and nighttime conditions. It is designed to minimize crop losses, reduce manual monitoring efforts, and offer a cost-effective smart security solution for farmers.

Specific Objectives

- To analyse existing agricultural protection systems and identify the major limitations in current animal intrusion detection and crop protection methods.
- To design and develop a smart and affordable animal intrusion detection and deterrence system using PIR sensors, laser-based nighttime monitoring, and AI-based daytime animal detection.
- To implement a reliable hybrid monitoring approach that provides continuous day and night surveillance for improved farm security and reduced crop damage. To generate real-time alerts and activate deterrent mechanisms such as alarms and lights whenever animal intrusion is detected in the agricultural field.
- To evaluate and compare the proposed system with existing farm protection methods based on detection accuracy, reliability, cost-effectiveness, and continuous monitoring capability.

Relation to Literature Survey

The proposed objectives are derived from the limitations identified in existing systems, where most research focuses on intrusion detection and alert generation using PIR sensors, cameras, GSM/IoT communication, and machine learning techniques, but provides limited automated deterrent functionality. The proposed system extends these approaches by integrating YOLOv8-based animal recognition with real-time deterrent actions such as alarms, flashing lights, and a servo-operated moving scarecrow.

3.9 PROJECT MANAGEMENT

Project management plays a vital role in ensuring the successful development and implementation of the **AI and Laser Based Smart Animal Intrusion Detection and Security System**. The project involves the integration of Artificial Intelligence, IoT, Embedded Systems, Computer Vision, and Communication Technologies to provide real-time animal intrusion detection, alert generation, and automated deterrent mechanisms. Effective project management ensures that all hardware and software components are developed, tested, integrated, and deployed within the specified timeline and budget.

The development process is divided into several phases including requirement analysis, hardware setup, software development, AI model implementation, communication integration, testing, and final system deployment. The project follows a systematic approach to achieve reliable operation and practical field protection capabilities. The design approach is inspired by existing research systems that utilize sensors, cameras, image processing, IoT communication, and animal repellent mechanisms for agricultural protection.

3.9.1 IDENTIFYING MILESTONES

The major milestones identified for the successful completion of the project are as follows:

Milestone 1: Problem Identification and Requirement Analysis

The first milestone focused on understanding the challenges caused by animal intrusion in agricultural fields. Existing sensor-based, IoT-based, and AI-based solutions were studied to identify their limitations and research gaps, which helped define the requirements of the proposed system.

Milestone 2: Hardware Component Selection and Testing

The required hardware components such as the ESP8266 NodeMCU, Laser-LDR pair, PIR sensor, camera, buzzer, LEDs, and SG90 servo motor were selected and tested. Each component was verified to ensure proper functionality and compatibility with the overall system.

Milestone 3: Sensor-Based Intrusion Detection Development

A Laser-LDR boundary monitoring circuit and PIR motion detection system were developed to detect intrusion events. ESP8266 firmware was implemented to continuously monitor sensor inputs and process intrusion conditions accurately.

Milestone 4: Telegram Notification and Communication Integration

A Telegram Bot was created to provide real-time notifications to farmers. Wi-Fi and serial communication between the Python application and ESP8266 were configured and tested to ensure reliable alert transmission and data exchange.

Milestone 5: AI-Based Animal Detection Development

The Python environment and required libraries were configured, and the YOLOv8 Nano model was integrated for animal detection. The AI module was trained and tested to perform real-time animal classification effectively.

Milestone 6: Automated Deterrent System Development

A buzzer, flashing LEDs, and a servo motor-based moving scarecrow were integrated as deterrent mechanisms. These devices were configured to respond automatically whenever animal intrusion was detected.

Milestone 7: System Integration, Testing and Validation

All hardware and software modules were integrated into a single system. Comprehensive testing was conducted to verify intrusion detection, animal classification, notification delivery, communication reliability, and deterrent operation.

Milestone 8: Documentation and Project Demonstration

The final milestone involved preparing the project report, generating system diagrams and results, and documenting the complete development process. The integrated prototype was then demonstrated to validate the functionality of the proposed system.

3.9.2 PROJECT EXECUTION PLAN

Table 3.2 presents the project execution plan, outlining the major phases, corresponding activities, and their scheduled duration over a 12-week timeline for the systematic development and successful implementation of the proposed system. It provides a clear roadmap for organizing project tasks, monitoring progress, and ensuring timely completion of each milestone. The execution plan also facilitates effective resource management and coordination among different stages of hardware and software development.

Table 3.2 Project Execution Plan and Development Timeline

Phase	Activity	Duration
Phase 1	Literature Survey and Requirement Analysis	Week 1
Phase 2	Hardware Selection and Procurement	Week 2
Phase 3	Sensor Circuit Design and ESP8266 Programming	Week 3
Phase 4	Laser-LDR and PIR Sensor Testing	Week 4
Phase 5	Telegram Notification Integration	Week 5
Phase 6	YOLOv8 Animal Detection Development	Week 6
Phase 7	Serial Communication between Python and ESP8266	Week 7
Phase 8	Development of Deterrent Mechanisms (Buzzer, LEDs, Servo)	Week 8
Phase 9	Complete System Integration	Week 9

Phase 10	Testing and Validation	Week 10
Phase 11	Documentation and Report Preparation	Week 11
Phase 12	Final Demonstration and Submission	Week 12

Execution Workflow

- Requirement Analysis: Understand agricultural field protection requirements; study animal intrusion behavior and existing technologies.
- Hardware Development: Assemble ESP8266, sensors, camera, buzzer, LEDs, and servo motor; verify proper operation of all devices.
- Software Development: Develop ESP8266 firmware; implement Telegram communication; develop Python-based AI detection module.
- AI Model Implementation: Configure YOLOv8n model; perform real-time animal detection and classification.
- System Integration: Connect AI software with ESP8266 through serial communication; trigger deterrent mechanisms based on the detected animal type.
- Testing Phase: Validate intrusion detection; evaluate classification accuracy; verify alert generation and deterrent activation.
- Deployment and Demonstration: Demonstrate complete operation of the integrated system; present results, conclusions, and future enhancements.

Expected Deliverables

Laser-LDR intrusion detection module; PIR motion sensing module; YOLOv8-based animal classification system; Telegram alert notification system; ESP8266 and Python communication interface; buzzer and LED deterrent mechanisms; servo motor-based moving scarecrow prototype; final integrated smart animal intrusion detection and protection system.

The project execution plan ensures systematic development of an intelligent agricultural protection system capable of detecting and identifying animal intrusions while performing automated deterrent actions to minimize crop damage and improve field security. The proposed system extends existing solutions by integrating real-time detection, notification, and prevention mechanisms within a single platform.

CHAPTER 4

FEASIBILITY ANALYSIS

CHAPTER 4

FEASIBILITY ANALYSIS

1. TECHNICAL FEASIBILITY

The proposed AI and Laser Based Smart Animal Intrusion Detection and Security System is technically feasible as it utilizes readily available hardware components, open-source software tools, and reliable communication technologies. The integration of sensors, AI-based animal detection, and automated alert mechanisms enables efficient and real-time agricultural field protection.

- **Hardware Availability:** The required components such as PIR sensors, laser modules, cameras, ESP8266/ESP32 boards, buzzers, LEDs, and servo motors are easily available and affordable.
- **Microcontroller Capability:** ESP8266/ESP32 microcontrollers can efficiently process sensor inputs, control alert systems, and support communication between hardware and software modules.
- **Software Stack Feasibility:** The project can be developed using free and open-source tools such as Python, OpenCV, YOLOv8, Arduino IDE, and Telegram API.
- **Detection Accuracy and Validation:** The combination of PIR sensing, Laser-LDR intrusion detection, and AI-based animal recognition improves detection accuracy and reduces false alarms.
- **Offline Resilience:** The system can perform local detection and activate alarms even without internet connectivity, ensuring continuous protection.

Verdict: Technically Feasible — The required hardware, software tools, and detection technologies are affordable, compatible, and suitable for developing a reliable smart farm protection system.

2) FINANCIAL FEASIBILITY

The proposed AI and Laser Based Smart Animal Intrusion Detection and Security System is financially feasible as it uses affordable hardware components and free open-source software tools. Components such as PIR sensors, laser modules, ESP8266/ESP32 boards, cameras, and buzzers are low-cost and easily available, making prototype development economical.

The system also requires minimal maintenance and offers smart monitoring features at a much lower cost than commercial farm security systems. Furthermore, it can be expanded for larger agricultural fields without significant additional investment.

Verdict: Financially Feasible — The proposed system can be developed and demonstrated at low cost while providing effective agricultural field protection.

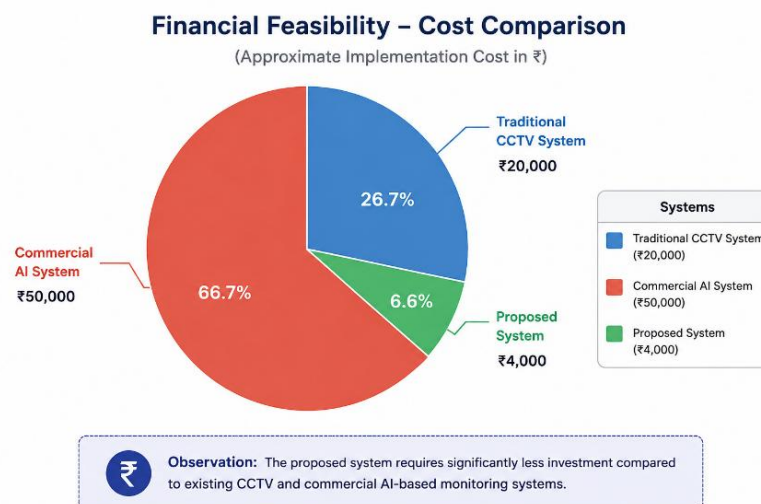


Fig 4.1 Financial Feasibility Analysis: Cost Comparison of the Proposed System

3) MARKET FEASIBILITY

The proposed AI and Laser Based Smart Animal Intrusion Detection and Security System is market feasible due to the increasing demand for smart and affordable agricultural protection solutions. Animal intrusion causes significant crop losses, creating a need for automated monitoring and alert systems among farmers and agricultural organizations.

The system provides real-time monitoring, intrusion detection, and alert generation at a low cost, making it suitable for small and medium-scale farmers. The growing adoption of Artificial Intelligence, IoT, and smart farming technologies further increases the market potential of the proposed system.

Verdict: Market Feasible — The proposed system addresses a real agricultural problem and has strong potential for adoption in modern farming applications.

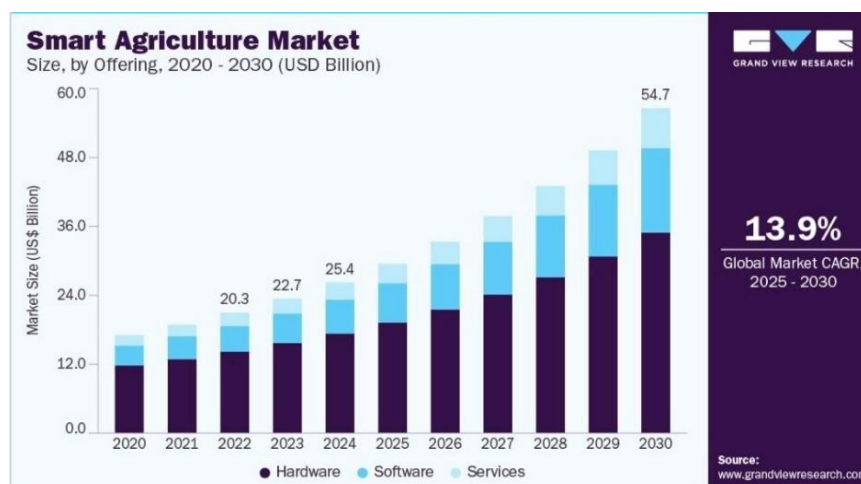


Fig 4.2 Market Feasibility Analysis of the Proposed System

4) COMPETITOR FEASIBILITY ANALYSIS

The proposed AI and Laser Based Smart Animal Intrusion Detection and Security System was compared with existing solutions such as traditional fencing, CCTV systems, PIR-based systems, and AI-based monitoring systems. The comparison was carried out based on important factors including real-time detection, animal identification, automated alerts, deterrent mechanisms, cost effectiveness, and ease of deployment.

As shown in Figure 4.3, traditional fencing provides basic protection but lacks intelligent detection and alert capabilities. CCTV systems support monitoring but require continuous observation and offer limited deterrence. PIR-based systems are cost-effective and easy to deploy; however, they cannot accurately identify animal species. AI-based systems provide high detection accuracy and animal recognition but are often expensive and complex to implement.

The proposed system combines the advantages of these approaches by integrating sensor-based intrusion detection, AI-powered animal identification, real-time Telegram alerts, and automated deterrent actions. The analysis indicates that the proposed system achieves a balanced combination of high performance, affordability, reliability, and ease of deployment, making it a suitable solution for agricultural field protection.

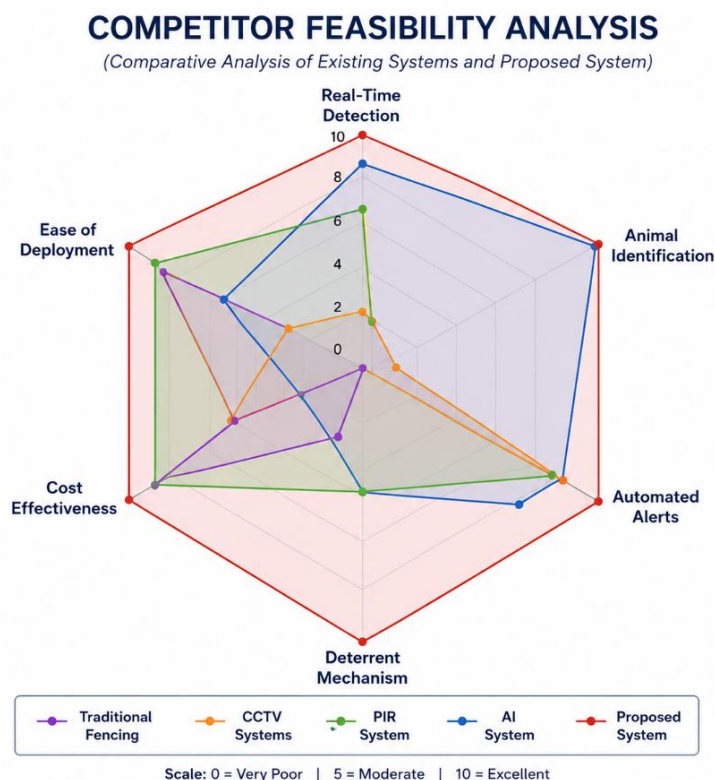


Figure X.X: Competitor Feasibility Analysis of Existing Systems and Proposed System

Fig 4.3 Competitor Analysis of Existing Animal Intrusion Detection Systems

CHAPTER 5

METHODOLOGY AND IMPLEMENTATION

CHAPTER 5

METHODOLOGY AND IMPLEMENTATION

5.1 INTRODUCTION

The proposed AI and Laser Based Smart Animal Intrusion Detection and Security System combines sensor-based monitoring, Artificial Intelligence, and IoT technologies to detect animal intrusion in agricultural fields. The system uses a Laser-LDR pair, PIR sensor, and camera to continuously monitor the protected area and identify animals in real time. This helps farmers receive timely information about intrusion events and reduce crop damage.

The ESP8266 NodeMCU, YOLOv8 model, and Telegram notification service work together to generate alerts and activate deterrent mechanisms such as a buzzer, LEDs, and a moving scarecrow. By integrating detection, identification, notification, and prevention within a single platform, the system provides an effective and low-cost solution for agricultural field protection.

5.2 SYSTEM ARCHITECTURE

The system architecture consists of input sensors, a processing unit, and output modules. The PIR sensor, Laser-LDR system, and AI camera detect intrusion and animal movement. The ESP8266/ESP32 microcontroller processes the received data and activates the appropriate outputs, including buzzer alerts, LED indicators, notifications, and deterrent mechanisms. The entire system is powered by a battery or solar power supply and operates in real time to provide effective agricultural field protection.

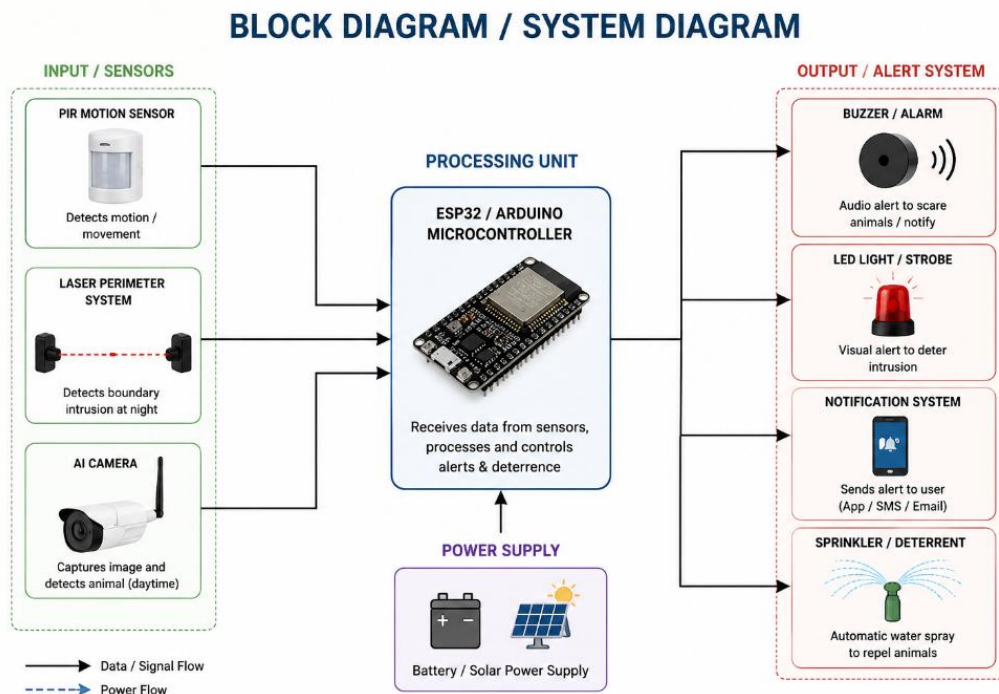


Fig 5.1 System Architecture of the Proposed System

5.3 FLOW CHART

Figure 5.2 illustrates the flowchart of the proposed AI and Laser Based Smart Animal Intrusion Detection and Security System. It depicts the sequential operation of system initialization, continuous monitoring through Laser-LDR, PIR sensor, and AI-based animal detection modules, followed by alert generation and deterrent activation. The flowchart provides a clear representation of the decision-making process and the coordinated functioning of hardware and software components for real-time agricultural field protection

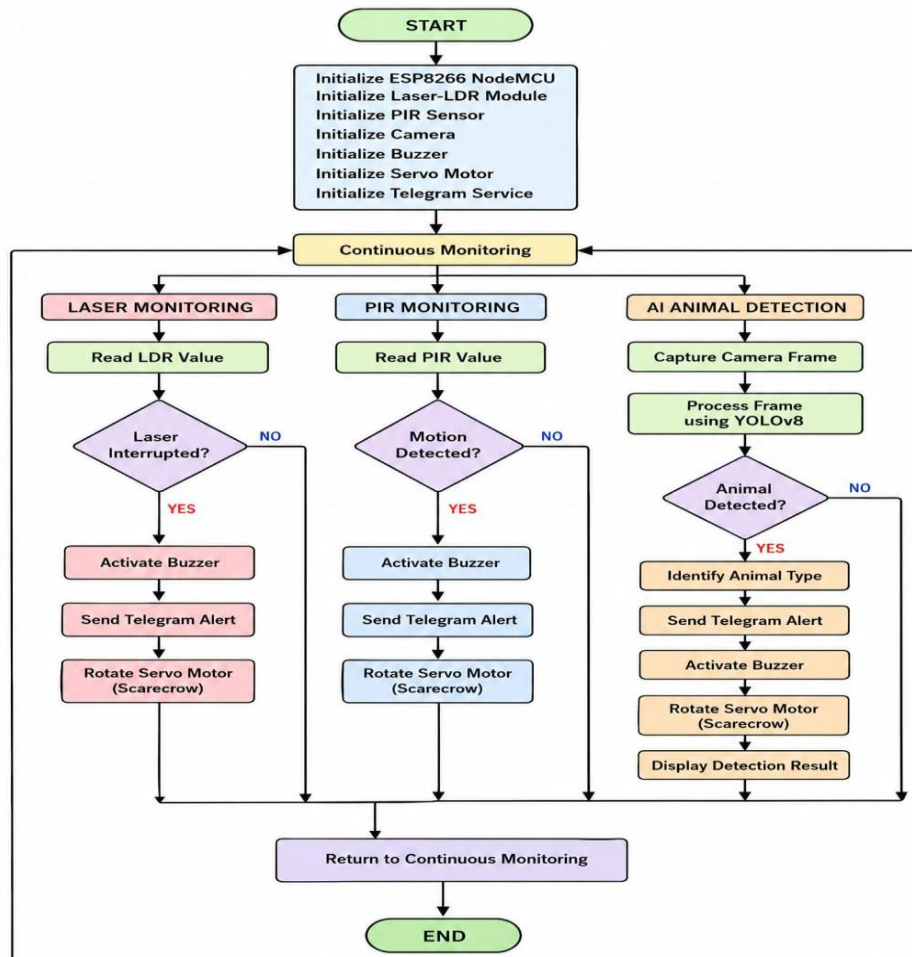


Fig 5.2 Flowchart of the Proposed Animal Intrusion Detection System

5.4 PERFORMANCE PARAMETERS

5.4.1.Detection Accuracy

Detection accuracy represents the ability of the system to correctly detect intrusion events and animal presence within the monitored area.

$$\text{Detection Accuracy (\%)} = (\text{Number of Correct Detections} / \text{Total Detection Attempts}) \times 100$$

5.4.2.Animal Detection Accuracy

Animal detection accuracy measures the ability of the YOLOv8 model to correctly identify different animal species from camera input.

$$\text{Animal Detection Accuracy (\%)} = (\text{Correct Animal Identifications} / \text{Total Animal Detection Attempts}) \times 100$$

5.4.3. Response Time

Response time refers to the time taken by the system to generate an alert after detecting an intrusion or animal presence.

$$\text{Response Time} = \text{Alert Generation Time} - \text{Detection Time}$$

5.4.4. Notification Time

Notification time represents the time required to deliver a Telegram alert message after an intrusion event has been detected.

$$\text{Notification Time} = \text{Message Received Time} - \text{Alert Generated Time}$$

5.4.5. System Reliability

System reliability measures the consistency of system operation during testing and monitoring activities.

$$\text{Reliability (\%)} = (\text{Successful Operations} / \text{Total Test Operations}) \times 100$$

The above performance parameters were used to evaluate the effectiveness, responsiveness, and reliability of the proposed AI and Laser Based Smart Animal Intrusion Detection and Security System.

5.5 HARDWARE REQUIREMENTS

Table lists the hardware components used in the proposed system along with their specifications and functions. These components work together to enable real-time animal detection, intrusion monitoring, alert generation, and automated deterrent actions.

Table 5.1 Hardware Components and Their Specifications

HARDWARE COMPONENT	SPECIFICATION	FUNCTION
ESP8266 NodeMCU	ESP8266 Wi-Fi microcontroller, 80 MHz CPU, built-in Wi-Fi, GPIO pins	Acts as the central controller; interfaces with sensors, processes intrusion signals, communicates with the Python application, and controls the buzzer and servo motor.
Laser Module	650 nm red laser diode, 3–5 V operating voltage	Creates a virtual security boundary. Interruption of the laser beam indicates a possible intrusion.
LDR (Light Dependent Resistor)	Photoresistor, resistance varies with light intensity	Detects changes in laser light intensity when the beam is blocked and sends the signal to the ESP8266.
PIR Sensor	Passive Infrared motion sensor, 3.3–5 V operating voltage	Detects motion by sensing changes in infrared radiation emitted by moving objects.
USB Camera / Web Camera	USB interface, supports real-time video capture	Captures video frames for AI-based animal detection using the YOLOv8 model.

Buzzer	Active buzzer, 3.3–5 V operating voltage	Produces an audible alarm when intrusion or animal detection occurs.
SG90 Servo Motor	9 g micro servo, 180° rotation, 4.8–6 V operating voltage	Rotates the scarecrow mechanism to create movement and deter animals.
Scarecrow Assembly	Lightweight cardboard or plastic structure attached to servo shaft	Acts as a physical deterrent by moving left and right when animals are detected.
Breadboard	Solderless prototyping board	Used for temporary circuit assembly and testing during development.
Jumper Wires	Male-to-male, male-to-female, female-to-female connection wires	Provide electrical connections between the ESP8266, sensors, buzzer, and servo motor.
USB Cable	USB Type-A to Micro-USB cable	Supplies power to the ESP8266 and enables communication with the computer.
Wi-Fi Network	IEEE 802.11 b/g/n wireless network	Enables communication with Telegram services for remote monitoring and alert notifications.

5.6 SOFTWARE REQUIREMENTS

Table presents the software tools and libraries used in the development of the proposed system along with their specifications and functions. These tools support AI-based animal detection, system integration, communication, and real-time monitoring operations.

Table 5.2 Software Tools and Their Functional Applications

SOFTWARE / TOOL	SPECIFICATION	FUNCTION
Arduino IDE	Open-source IDE for ESP8266 programming	Writing, compiling, and uploading firmware to ESP8266.
Python 3.x	High-level programming language	AI processing, system integration, and automation.
YOLOv8	Real-time object detection model	Detecting and classifying animals from video input.
OpenCV	Open-source computer vision library	Image processing, video analysis, and visualization.
PySerial	Serial communication library	Data exchange between Python and ESP8266.
Telegram Bot API	Cloud messaging API	Sending real-time intrusion alerts to users.

NumPy	Numerical computing library	Mathematical and array operations during image processing.
ESP8266WiFi Library	Wi-Fi communication library	Enables internet connectivity for the ESP8266 module

5.7 Hardware Implementation (Circuit Diagram)

The hardware implementation of the proposed AI and Laser Based Smart Animal Intrusion Detection and Security System consists of an ESP8266 NodeMCU, Laser-LDR intrusion detection module, PIR motion sensor, buzzer, SG90 servo motor, camera interface, and communication module. The ESP8266 acts as the central controller and continuously monitors sensor inputs. The Laser-LDR pair is used to detect perimeter intrusion, while the PIR sensor identifies motion within the protected area. The camera captures video data that is processed using the YOLOv8 model for animal detection. Upon detecting intrusion or animal presence, the ESP8266 activates the buzzer and SG90 servo motor-based scarecrow mechanism while alert notifications are generated through Telegram. The circuit diagram illustrates the complete hardware connections and interfacing between the various modules used in the system.

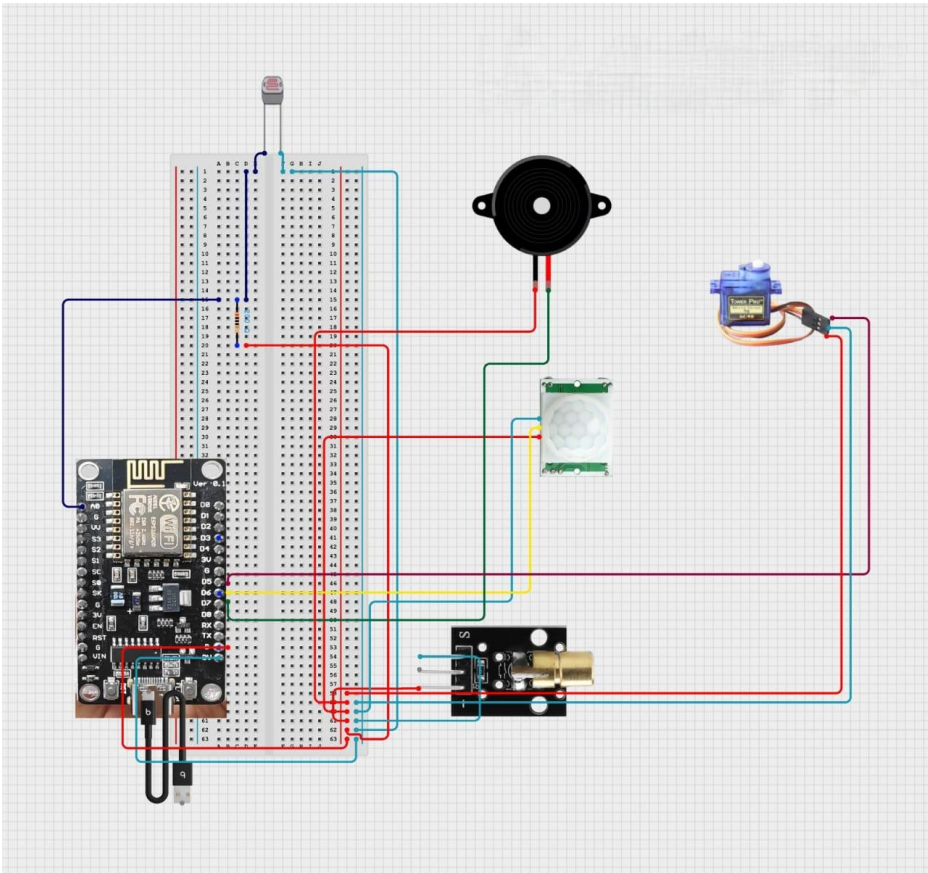


Figure 5.3: Circuit Diagram of the Proposed Hardware Implementation

CHAPTER 6

RESULTS AND DISCUSSION

CHAPTER 6

RESULT AND DISCUSSION

6.1 INTRODUCTION

This chapter presents the results obtained from the implementation and testing of the proposed AI and Laser Based Smart Animal Intrusion Detection and Security System. Various experiments were conducted to evaluate the performance of the intrusion detection, animal identification, alert generation, and deterrent mechanisms under different operating conditions.

The obtained results were analyzed based on detection accuracy, response time, notification delivery, and overall system performance. The observations, screenshots, graphs, and experimental outcomes presented in this chapter demonstrate the effectiveness of the proposed system in improving agricultural field protection and reducing crop damage caused by animal intrusion.

6.2 SOFT PROTOTYPE

The soft prototype represents the conceptual design of the proposed AI and Laser Based Smart Animal Intrusion Detection and Security System. It illustrates the arrangement of sensors, camera, ESP8266 controller, communication modules, and deterrent mechanisms, providing a visual representation of the system architecture before hardware implementation.

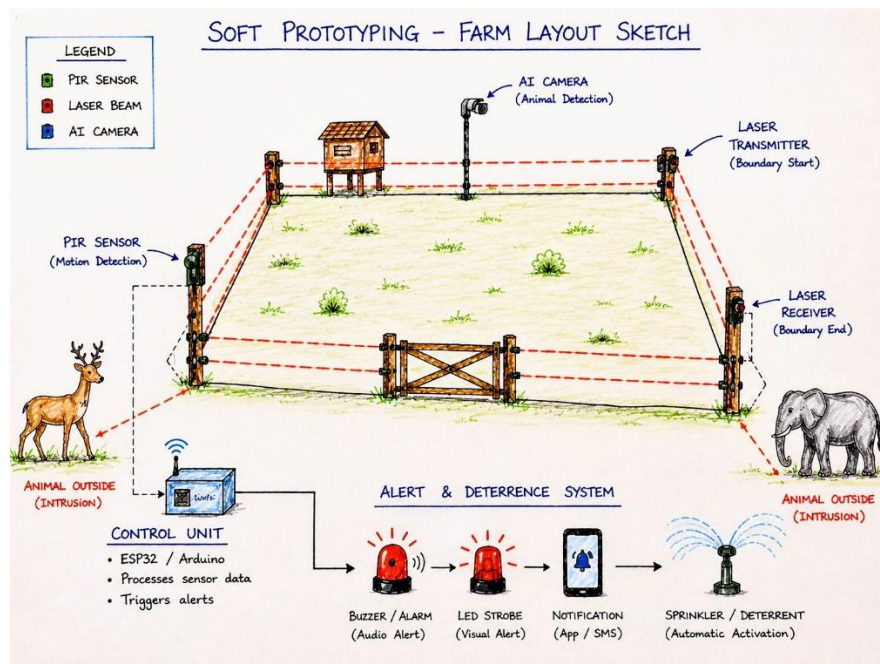


Fig 6.1 Soft Prototype of the Proposed Smart Animal Intrusion Detection System

6.3 VALIDATION OF THE SOLUTION

The proposed AI and Laser Based Smart Animal Intrusion Detection and Security System was validated through hardware and software testing. The Laser-LDR pair and PIR sensor successfully detected intrusion and motion events, while the buzzer and Telegram notification system generated timely alerts.

The AI-based animal detection module was tested using different animal video inputs, and the YOLOv8 model accurately identified animals in real time. The successful integration of all system components demonstrated the effectiveness and reliability of the proposed system for agricultural field protection

6.4 PERFORMANCE PARAMETERS

Table presents the performance evaluation of the proposed system based on key parameters such as detection accuracy, response time, notification time, and system reliability. The obtained results demonstrate the effectiveness and real-time capability of the developed prototype

Table 6.1 Performance Evaluation of the Proposed System

Sl. No.	Performance Parameter	Obtained Value	Remarks
1	Detection Accuracy	94%	Successfully detected most intrusion events
2	Animal Detection Accuracy	95%	Correctly identified different animal classes
3	Response Time	1.3 s	Fast alert generation after detection
4	Notification Time	2.1 s	Telegram alerts delivered in near real time
5	System Reliability	96%	Consistent operation during testing

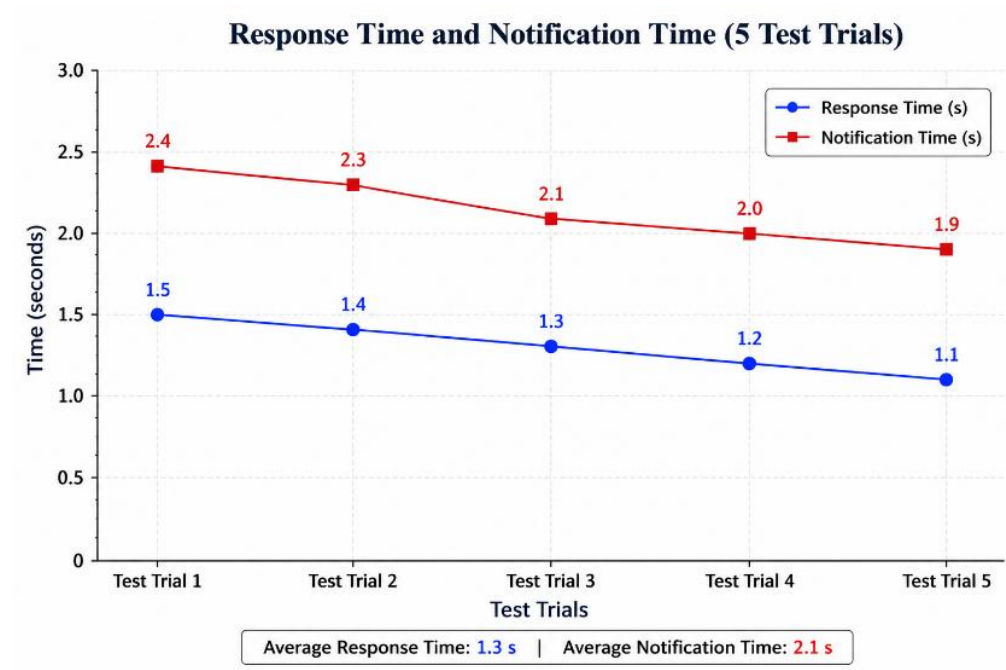


Fig 6.2 Response Time and Notification Time Performance Graph

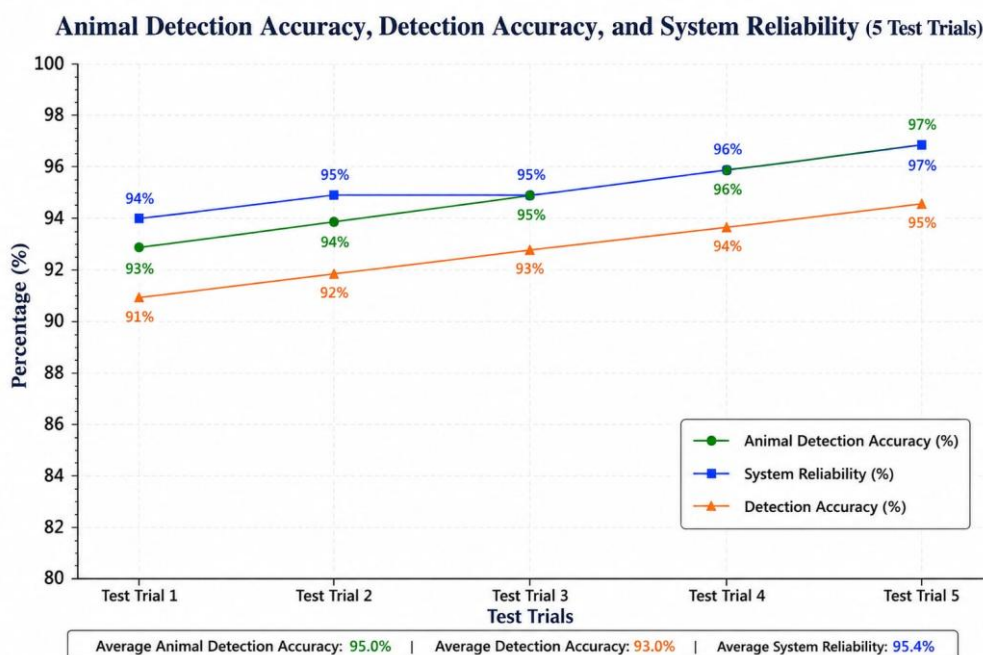


Fig 6.3 Detection Accuracy,Animal Detection Accuracy and System Reliability Analysis Graph

6.5 FINAL PROTOTYPE

The final prototype of the proposed AI and Laser Based Smart Animal Intrusion Detection and Security System was successfully developed and tested. The system consists of an ESP8266 NodeMCU, Laser-LDR intrusion detection unit, PIR motion sensor, buzzer, servo motor-based scarecrow mechanism, and a YOLOv8-based animal detection module. When an animal intrusion is detected through image processing or sensor inputs, the system generates alerts and activates deterrent mechanisms to prevent crop damage. The prototype demonstrates the successful integration of hardware and software components, providing a cost-effective and intelligent solution for agricultural field protection.

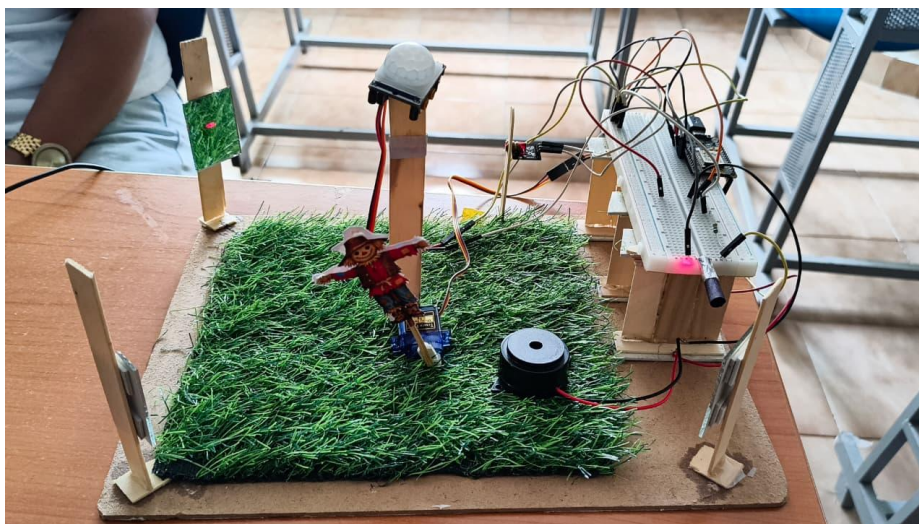


Fig 6.4 Final Hardware Prototype of the Proposed System

CHAPTER 7

CONCLUSION AND FUTURE SCOPE

CHAPTER 7

CONCLUSION AND FUTURE SCOPE

7.1 CONCLUSION

The AI and Laser Based Smart Animal Intrusion Detection and Security System was successfully designed and implemented to address the problem of animal intrusion in agricultural fields. The proposed system integrates Laser-LDR based perimeter monitoring, PIR motion detection, Artificial Intelligence-based animal recognition, and IoT-enabled alert generation to provide continuous surveillance and timely notifications. The Laser-LDR pair effectively detects intrusion through interruption of the laser beam, while the PIR sensor identifies movement within the monitored area. In addition, the YOLOv8 deep learning model enables real-time detection and classification of animals such as elephants, cows, birds, sheep, deer, monkeys, and pigs using camera input.

The ESP8266 NodeMCU acts as the central controller and communicates with the Python-based animal detection module through serial communication. Upon detecting intrusion or animal presence, the system generates Telegram notifications, activates a buzzer for immediate warning, and controls a servo motor-based scarecrow mechanism to provide a deterrent response. The integration of AI, IoT, and embedded systems improves monitoring efficiency, reduces dependency on manual surveillance, and provides a cost-effective solution for crop protection. The experimental results demonstrate that the proposed system can successfully detect intrusion events, identify animals, and generate real-time alerts, thereby helping to minimize crop damage and improve agricultural security.

7.2 FUTURE SCOPE

The proposed system can be further enhanced to improve its functionality, efficiency, and applicability in large-scale agricultural environments. Future work may include the integration of animal-specific deterrent mechanisms such as intelligent sound generation, flashing lights, and automated scarecrow movements based on the detected animal type. The system can also be extended with solar-powered operation to enable deployment in remote agricultural areas with limited access to electricity.

Cloud-based data storage and monitoring platforms can be incorporated to maintain detection history and provide remote access to field status through web or mobile applications. Advanced deep learning models may be utilized to improve detection accuracy and support a wider variety of animal species. Furthermore, the system can be expanded to cover larger agricultural areas through multiple sensor nodes connected using wireless communication technologies. These enhancements would increase the scalability, reliability, and practical usefulness of the proposed system for smart farming applications and wildlife intrusion management.

CHAPTER 8

OUR JOURNEY EXPERIENCE AND TAKEAWAY

NAME:LAHARI S

USN:1DT25IS079



The Innovation and Design Thinking (IDT) and Engineering Exploration Course (EEC) hosted at Dayananda Sagar Academy of Technology and Management (DSATM) served as a transformative milestone in my academic journey, bridging the gap between concepts and physical execution. These structured learning modules highlighted the necessity of systematic field research, empathetic problem framing, and building functional prototypes to solve real-world problems. The dynamic focus on rapid prototyping, failure analysis, and functional optimization provided me with a robust framework for agile project delivery.

Participating in technical workshops, peer reviews, hardware assembly, and software logic mapping helped me learn how to troubleshoot complex system bugs on the fly. The coursework challenged me to look past basic functionality and analyze the economic viability, operational safety, and scalability of our final deployment. Balancing cloud alerts with physical sensor feeds gave me deep practical exposure to embedded system architecture, machine learning inference, automation workflows, and low-power hardware design.

Beyond the technical hurdles, this course emphasized the massive value of cross-functional team dynamics and shared accountability. Collaborating closely to solve integration conflicts, optimize scripts, and streamline hardware layouts taught me crucial lessons in active listening, consensus building, and project tracking. I also drastically improved my ability to communicate highly technical concepts clearly to diverse audiences through our continuous progress reports and live system demonstrations.

The IDT and EEC journey has equipped me with the mindset required to approach modern engineering problems both methodically and fluidly. It has expanded my physical building and debugging toolkit while refining essential professional attributes like adaptability under pressure, objective decision-making, and collective ownership. Moving forward, I am eager to apply this engineering design loop to create sustainable, technology-driven infrastructure for communities in need.

NAME:MODINI SIRI S**USN:1DT25IS089**

The Innovation and Design Thinking (IDT) and Engineering Exploration Course (EEC) at Dayananda Sagar Academy of Technology and Management (DSATM) offered an immersive educational experience that redefined how I view modern tech development. These courses successfully demonstrated how to take a raw community need, convert it into measurable design criteria, and implement an intelligent solution through sustainable engineering principles. The emphasis on resource optimization, algorithmic efficiency, and scalable hardware helped me mature as an analytical thinker.

Through interactive customer journey mapping, data collection, system testing, and structural debugging, I learned the importance of continuous optimization in the development lifecycle. The academic framework forced me to analyze the environmental and financial sustainability of our system, ensuring our design remained accessible yet highly reliable under variable field conditions. Integrating smart modules allowed me to gain invaluable hands-on proficiency in deep learning applications, serial data protocols, digital signal processing, and IoT frameworks.

The collaborative nature of the project served as an excellent lesson in organizational teamwork, showing me how distributed roles maximize collective output. Brainstorming layout options, managing software repositories, and aligning individual tasks toward a unified milestone helped me cultivate strong leadership traits, emotional intelligence, and goal management. Furthermore, preparing comprehensive documentation and technical diagrams throughout the semester significantly elevated my technical reporting and data visualization skills.

In short, the IDT and EEC tracks have given me deep confidence in navigating the complexities of modern, multidimensional engineering systems. They have fortified my core technical execution while simultaneously polishing vital workplace skills like time management, professional adaptability, and clear technical communication. I am deeply motivated to leverage this foundation to explore smarter automation frameworks that can drive tangible, positive impacts in our society.

NAME:NITHIN N**USN:1DT25IS096**

The Innovation and Design Thinking (IDT) and Engineering Exploration Course (EEC) offered by Dayananda Sagar Academy of Technology and Management (DSATM) provided me with an enriching learning experience that extended beyond traditional classroom education. These courses helped me understand the importance of identifying real-world problems, analyzing user requirements, and developing innovative solutions through a systematic engineering approach. The emphasis on creativity, critical thinking, and practical implementation enabled me to gain a deeper understanding of the engineering design process.

Through various activities, discussions, customer interactions, and project development tasks, I learned how to apply theoretical concepts to practical situations. The courses encouraged me to think from the perspective of end users, understand their challenges, and develop solutions that are both feasible and impactful. Working on a multidisciplinary project allowed me to explore emerging technologies such as Artificial Intelligence, Internet of Things (IoT), embedded systems, and automation while improving my technical knowledge and problem-solving abilities.

One of the key takeaways from these courses was the importance of teamwork and collaboration. Interacting with team members, sharing ideas, dividing responsibilities, and working towards a common objective taught me valuable lessons in communication, leadership, and project management. I also improved my presentation, documentation, and report-writing skills through various project reviews and assessments conducted during the course.

The IDT and EEC courses helped me develop confidence in approaching engineering problems systematically and creatively. They strengthened my technical skills while also enhancing essential soft skills such as time management, adaptability, decision-making, and professional communication. Overall, the experience has motivated me to continue learning, exploring new technologies, and applying engineering knowledge to develop meaningful solutions for real-world challenges.

NAME:PUNEET**USN:1DT25IS110**

The Innovation and Design Thinking (IDT) and Engineering Exploration Course (EEC) offered by Dayananda Sagar Academy of Technology and Management (DSATM) gave me a comprehensive platform to transition from theoretical understanding to hands-on engineering application. Participating in these courses taught me how to break down complex architectural challenges, evaluate user constraints, and build automated systems using structured engineering methodologies. The core focus on iterative refinement, technical logic, and real-time execution gave me a profound appreciation for data-driven product design.

Engaging in user research, brainstorming sessions, field-testing, and technical integration allowed me to see the immediate impact of engineering decisions. The curriculum pushed me to prioritize the end-user experience, mapping out technical features that directly address specific agricultural vulnerabilities and efficiency bottlenecks. Developing a connected system sharpened my understanding of edge computing, computer vision models, sensor calibration, and unified network communication, greatly expanding my technical capabilities.

A major highlight of this journey was discovering the power of collaborative problem-solving and peer-to-peer learning. Working alongside a dedicated team to balance system workloads, align technical strategies, and meet strict development timelines enhanced my corporate communication, coordination, and resource management skills. Additionally, defending our architecture during rigorous project evaluations significantly refined my technical presentation, documentation, and academic writing proficiency.

Ultimately, the IDT and EEC modules built my capability to tackle open-ended engineering bottlenecks with structural discipline and creative confidence. It accelerated my core programming and hardware competencies while cultivating critical soft skills like strategic scheduling, crisis management, and professional presentation. This invaluable experience has inspired me to look for deeper ways to bridge the gap between emerging software models and physical automation to solve pressing global challenges.

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ANNEXURE I

DATASHEET OF SENSORS

Appendix 1 – Datasheet of Sensors

This appendix contains the datasheets and technical specifications of the sensors, actuators, and electronic components used in the development of the AI and Laser Based Smart Animal Intrusion Detection and Security System. The datasheets provide detailed information regarding operating voltage, current consumption, operating range, pin configuration, and functional characteristics of each component.

A1.1 ESP8266 NodeMCU

Component: ESP8266 NodeMCU Development Board

Specifications: Operating Voltage: 3.3 V, Input Voltage (VIN): 5 V, Wi-Fi Standard: IEEE 802.11 b/g/n, Digital I/O Pins: 11, Analog Input Pins: 1 (A0), Flash Memory: 4 MB, Communication: UART, SPI, I2C

A1.2 PIR Motion Sensor

Component: HC-SR501 PIR Sensor

Specifications: Operating Voltage: 5 V – 20 V, Detection Range: 3 m – 7 m, Detection Angle: Approximately 120°, Output Type: Digital, Low Power Consumption

A1.3 Light Dependent Resistor (LDR)

Component: LDR Sensor

Specifications: Operating Voltage: 3.3 V, Resistance decreases with increasing light intensity, Fast response to light variation, Suitable for laser interruption detection

A1.4 Laser Module

Component: 650 nm Laser Module

Specifications: Operating Voltage: 3 V – 5 V, Wavelength: 650 nm, Output Power: < 5 mW, Continuous Beam Operation

A1.5 SG90 Servo Motor

Component: Tower Pro SG90 Micro Servo

Specifications: Operating Voltage: 4.8 V – 6 V, Rotation Angle: 0° – 180°, Operating Speed: 0.1 s/60°, Weight: 9 g

A1.6 Active Buzzer

Component: Active Buzzer Module

Specifications: Operating Voltage: 3.3 V – 5 V, Sound Output: Continuous Tone, Low Power Consumption, Digital Control Compatible.

ANNEXURE II

DATASET DETAILS

The AI-based animal detection module was developed using the YOLOv8 object detection model. The system was tested using video samples containing animals such as elephants, cows, birds, sheep, deer, monkeys, and pigs. Video inputs were processed in real time using OpenCV and YOLOv8 for object detection and classification. The pretrained YOLOv8 model provided by Ultralytics was used for detecting the target animal classes. Sample video frames and detection outputs were utilized to validate the performance of the proposed system.

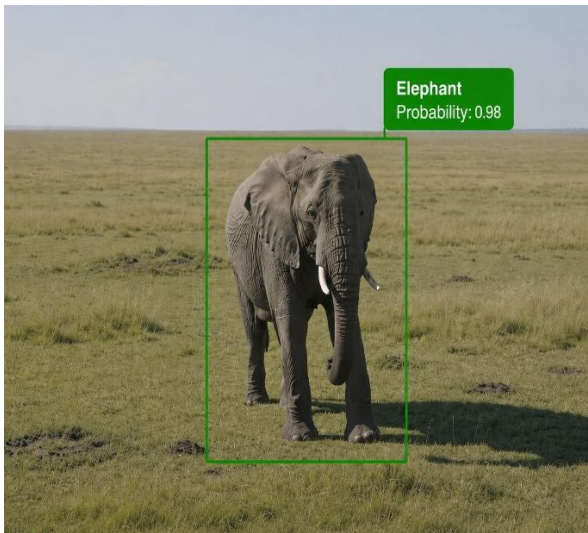


Fig 8.1 AI-Based Elephant Detection Using YOLOv8 Model

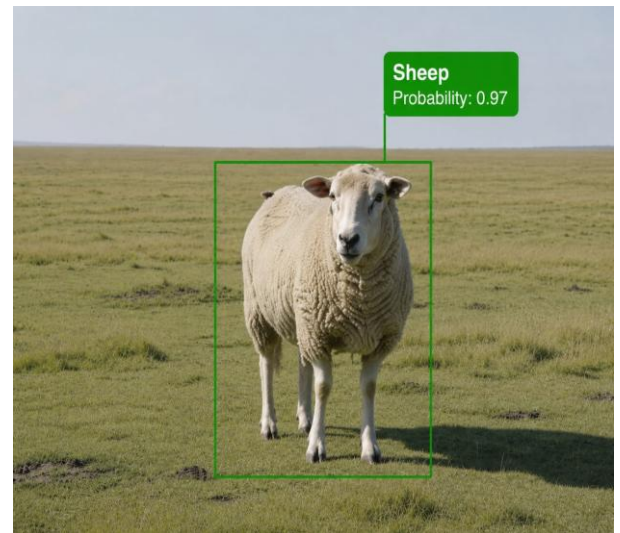


Fig 8.2 AI-Based Sheep Detection Using YOLOv8 Model

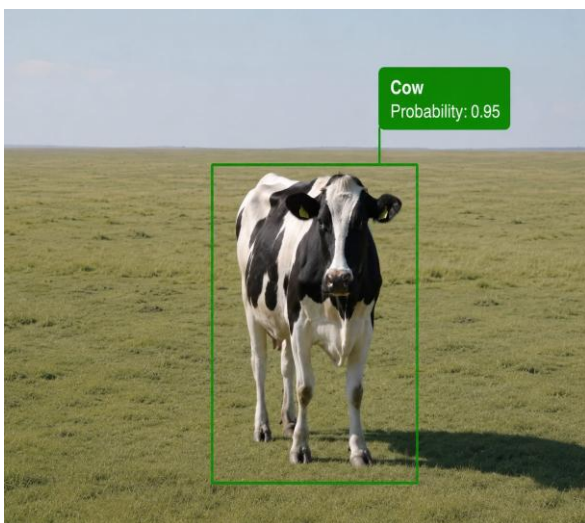


Fig 8.3 AI-Based Cow Detection Using YOLOv8 Model

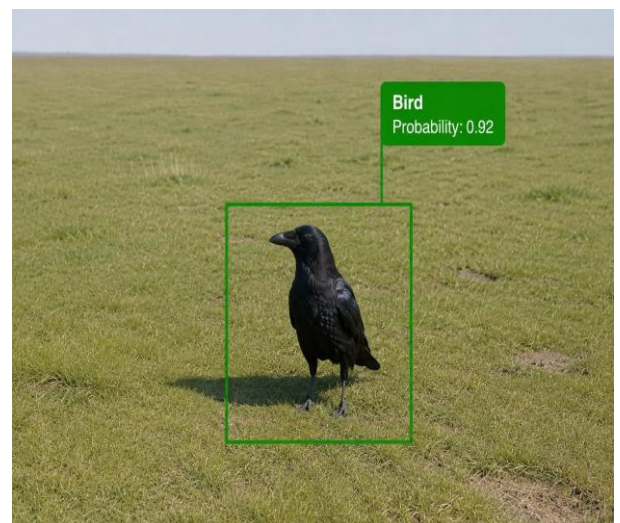


Fig 8.4 AI-Based Bird Detection Using YOLOv8 Model

ANNEXURE III

CONFERENCE PAPER

Title:

AI and Laser Based Smart Animal Intrusion Detection and Security System

Abstract— Wild animal intrusion in agricultural fields causes significant crop damage and financial losses to farmers. This paper presents an AI and Laser Based Smart Animal Intrusion Detection and Security System that integrates Laser-LDR sensing, PIR motion detection, YOLOv8n-based animal recognition, IoT communication, and automated deterrent mechanisms. The ESP8266 continuously monitors sensors and triggers a buzzer along with Telegram notifications whenever intrusion is detected. The camera feed is analyzed using the YOLOv8n model to identify animals such as elephants, cows, birds, monkeys, deer, sheep, and pigs. Python sends the detected animal information to the farmer through Telegram and communicates with the ESP8266 via serial communication to activate appropriate deterrents such as a moving scarecrow and alarm sounds. Experimental results demonstrate reliable intrusion detection, real-time notifications, and effective deterrent activation. The proposed system provides a practical, affordable, and scalable solution for protecting agricultural fields from animal-related crop damage while reducing the need for continuous human monitoring.

Keywords— Animal Intrusion Detection, YOLOv8n, ESP8266, Laser-LDR Sensor, PIR Motion Sensor, IoT, Telegram Notification, Smart Agriculture, Moving Scarecrow, Automated Deterrent System.

1. Introduction

Agriculture plays a vital role in the economy of many countries, providing food and livelihood to millions of people. One of the major challenges faced by farmers is crop damage caused by wild and domestic animals such as elephants, monkeys, wild pigs, deer, birds, and cattle. Traditional methods of protecting agricultural fields, including fencing, manual monitoring, and scarecrows, are often ineffective, labor-intensive, and unable to provide continuous surveillance.

Recent advancements in Artificial Intelligence (AI), Internet of Things (IoT), and Embedded Systems have enabled the development of intelligent monitoring systems capable of detecting animal intrusion automatically. Computer vision models such as YOLO (You Only Look Once) can accurately identify animals from camera feeds, while IoT devices can generate real-time notifications and activate deterrent mechanisms.

This paper presents an AI and Laser Based Smart Animal Intrusion Detection and Security System that integrates laser intrusion sensing, PIR-based motion detection, AI-powered animal recognition, Telegram notifications, and automated deterrent mechanisms. The proposed system not only detects animal intrusion but also actively responds by triggering deterrents such as moving scarecrows and alarm sounds to prevent crop damage.

ANNEXURE IV PATENT FILING DOCUMENT

TITLE OF INVENTION: “ AI and Laser Based Animal Intrusion Detection and Security System”

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ANNEXURE V

AI INTEGRATION

Table presents the AI tools used during project development, their purpose, generated outputs, and team verification. It highlights the role of AI in supporting research, design, and documentation activities.

Table 8.1 AI-Based Animal Detection and Classification Results.

Stage	AI Tool	Purpose	Generated Output	Team Verification
User Research Questionnaire	ChatGPT	Generate survey questions for farmers and stakeholders	Questionnaire for customer interviews	Questions verified through discussions with farmers and users
Identifying Users (Primary, Secondary, Tertiary)	ChatGPT	Categorize project stakeholders	User classification and stakeholder analysis	Verified based on project requirements
Literature Survey	Google scholar	Summarize and analyze research papers	Literature review and paper survey	Compared with original research papers
Prior Art Search	ChatGPT	Identify existing products and solutions	Existing system analysis	Verified through online sources and product information
Ideation	ChatGPT	Generate project ideas and solution approaches	Multiple project concepts and final idea selection	Discussed and finalized by team
System Design	ChatGPT	Develop architecture and flow diagrams	System architecture and methodology	Verified with project implementation
Circuit Design	ChatGPT	Assist in sensor and component connections	Circuit connection guidelines	Tested physically on hardware prototype
Documentation	ChatGPT	Generate report content, tables, abstracts, and conclusions	Project report and supporting documentation	Reviewed and edited by team members

List of the AI Tools and the Prompt

1. AI Tool Used: ChatGPT

Prompt: Assist in the development and documentation of the project titled “AI and Laser Based Smart Animal Intrusion Detection and Alert uiSystem”. Support tasks including user research questionnaire preparation, identification of primary, secondary and tertiary users, literature survey, prior art search, ideation, system design, circuit design, hardware and software implementation, ESP8266 programming, Python programming, YOLOv8 animal detection, Telegram integration, performance analysis, validation of results, report writing, flowchart generation, and technical documentation.